Two-Nonce Roster-Based Schnorr Signatures with Hidden Threshold Authorization

Protocol Specification, Security Assumptions, and Conditional Security Proofs

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Abstract

We specify and analyze a two-round, roster-based threshold Schnorr signature protocol in prime-order cyclic groups in which a hidden subset $S \subseteq [n]$ of at least t registered participants authorizes a freshly masked aggregate verification key $Q_e = \sum_{i \in S} PK_i + \xi G$. Its noninteractive authorization certificate π_{Auth} has linear size $O(n)$ and enforces four algebraic checks: commitment opening, exact public-key linkage, evaluated inner product, and threshold-polynomial consistency. We establish four conditional security results: (1) unconditional algebraic completeness; (2) straight-line threshold knowledge soundness and exact public-key linkage in a first-evaluation representation-annotated Algebraic Group Model with Random Oracles ($\mathsf{AGM} + \mathsf{ROM}$); (3) passive external-verifier signer-subset privacy in the programmable random-oracle model via an efficient polynomial-time hybrid simulator; and (4) concurrent unauthorized unforgeability reduced to the $(Q_{\text{sign}} + 1)$-Algebraic One-More Discrete Logarithm (AOMDL) and the d-generator Discrete Logarithm Relation (DLR) problems (with $\mathsf{Adv}_{\mathsf{DLR}} \leq d\,\mathsf{Adv}_{\mathsf{DLP}}$ under random generators). We prove both message-level threshold unforgeability ($|C \cup \mathsf{Resp}(m^*)| < t$) and subset-specific extracted authorization (no valid forgery can yield an extracted authorized subset S^* containing an honest participant $i^* \in S^* \setminus C$ who did not release a Round-2 response on m^*). All claims are explicitly scoped to static corruption with ordered extractable key registration and immutable first-evaluation public-basis annotations with composite honest-signer propagation.

1 Introduction and Technical Overview

Multi-party and threshold Schnorr signatures [1, 7, 5] allow a committee of n participants to jointly authorize messages while producing signatures verifiable against a public key. In decentralized governance, custody, and ad-hoc committee validation, three requirements frequently arise together:

(1) *DKG-free roster registration:* Participants register individual Schnorr keys $(PK_1, \ldots, PK_n)$ on a public roster without executing an interactive Distributed Key Generation (DKG) ceremony when the threshold t or committee is formed.

(2) *Public threshold verification with passive subset privacy:* Any subset $S \subseteq [n]$ of size $|S| \geq t$ signs in two rounds, while an external observer seeing only the final signature σ learns nothing about which subset S participated beyond $|S| \geq t$.

(3) *Concurrent unforgeability without rewinding blowup:* Security holds against an adversarial coordinator that interleaves and abandons concurrent signing sessions, avoiding ROS vulnerabilities [6] and multi-round rewinding loss.

While pairing-based accountable subgroup multisignatures [2, 8] require pairing-friendly curves and FROST [5] requires an interactive DKG for a fixed shared key, applying Bulletproofs logarithmic inner-product compression [4] to prove a t-of-n subset inside a masked key $Q_e = \sum_{i \in S} PK_i + \xi G$ requires $\Theta(\log n)$-round tree rewinding that is incompatible with straight-line AOMDL reductions.

Summary of contributions. We resolve this tension by pairing a two-nonce interactive Schnorr signing state machine (Algorithm 1) with an uncollapsed, linear-size ($O(n)$) polynomial authorization certificate π_{Auth} (Algorithm 2):

- **Straight-Line Threshold Extraction and Exact Linkage (Section 6):** Keeping the response vectors $v_L, v_R \in \mathbb{F}_q^N$ uncollapsed enables simultaneous verification of commitment opening (C1) against $(H_0, \mathbf{G}, \mathbf{H})$ and public-key linkage (C2) against PK_{pad}, yielding straight-line extraction in the first-evaluation annotated $\mathsf{AGM} + \mathsf{ROM}$ (Theorem 6.7).
- **Constructive Passive Signer-Subset Privacy (Section 7):** We prove that Q_e, R, s_e, and π_{Auth} reveal no information about the signing subset S to a passive external verifier via an explicit polynomial-time hybrid simulator (Theorem 7.5).
- **Message-Level and Subset-Specific Concurrent Unforgeability (Section 8):** Using a Three-World Distributional Coupling (Lemma 8.4), an exhaustive 4-partition Chronological Rank Lemma with deferred random-oracle sampling (Lemma 8.5), and an Opaque-Atom Affine Embedding (Lemma 8.6), we reduce concurrent unforgeability to $(Q_{\text{sign}} + 1)$-AOMDL and d-DLR / DLP (Theorem 8.8). Crucially, our rank analysis establishes both *message-level threshold unforgeability* ($|C \cup \mathsf{Resp}(m^*)| < t$) and *subset-specific authorization* ($S^* \setminus (C \cup \mathsf{Resp}(m^*)) \neq \emptyset$): an adversary cannot produce a valid signature whose extracted witness b^* attributes participation to any honest signer $i^* \in S^* \setminus C$ who did not respond on m^*.

Scheme	Setup	Group	Rounds	Subset Privacy	Signature Size	Concurrent Security
MuSig2 [7]	Roster (t=n)	Prime $\mathbb{G}$	2	N/A (t=n)	$2\mathbb{G}$ or $1\mathbb{G}+1\mathbb{F}_q$	AOMDL (ROM)
FROST [5]	Interactive DKG	Prime $\mathbb{G}$	2	Key-level	$1\mathbb{G}+1\mathbb{F}_q$	Threshold unforgeable
BLS ASM [8]	Roster ($1 \le t \le n$)	Pairing	1	None (S public)	$O(1)\mathbb{G}+n$ bits	Co-CDH (ROM)
This Work	**Roster** ($1 \le t \le n$)	**Prime** $\mathbb{G}$	**2**	**Passive (ROM)**	$7\mathbb{G}+(2N+4)\mathbb{F}_q$	$(Q_{\text{sign}}+1)$-AOMDL+DLR

Table 1: Comparison with representative multi-signature and threshold signature constructions.

2 Notation, Parameters, and Idealized Interfaces

2.1 Group, Dimensions, and Encodings

Let $\mathbb{G} = \langle G \rangle$ be an additively written cyclic group of prime order q, with identity element $\mathcal{O}$, and let $\mathbb{F}_q = \mathbb{Z}/q\mathbb{Z}$. Scalars and group elements are encoded canonically. The encoding function enc is injective and prefix-free on typed tuples, with a distinct domain-separation tag for each protocol purpose. Point validation enforces membership in $\mathbb{G}$ and canonical encoding; the identity element $\mathcal{O}$ is rejected exactly where the algorithm specification requires a nonidentity point. In particular, intermediate authorization proof commitments $A, S_{\text{com}}, Q_s, T_1, T_2$ are permitted to equal $\mathcal{O}$.

Fix roster size $n \ge 1$ and signing threshold $1 \le t \le n$, and define the dimensions

$$l = \lceil \log_2(n+1) \rceil, \qquad N = n + l + 1, \qquad d = 2N + 1. \tag{1}$$

For the field-to-integer threshold lifting in the authorization extraction theorem, assume

$$q > \max\{\, N + 1,\ n + 2^l - 1,\ 3 \,\}. \tag{2}$$

Condition (2) guarantees that a bounded integer threshold relation holding modulo q holds identically over $\mathbb{Z}$.

For vectors $u, v \in \mathbb{F}_q^N$ and a tuple of group elements $\mathbf{P} = (P_1, \ldots, P_N) \in \mathbb{G}^N$, we write $\langle u, v \rangle = \sum_{i=1}^N u_i v_i \pmod q$, $\langle u, \mathbf{P} \rangle = \sum_{i=1}^N u_i P_i \in \mathbb{G}$, and $(u \circ v)_i = u_i v_i$ for the entrywise

Hadamard product. For a finite set S, $|S|$ denotes its cardinality, and $a \leftarrow S$ denotes independent uniform sampling from S. Vectors of length N are indexed $1, \ldots, N$, whereas the l-bit surplus vector $d_s \in \{0,1\}^l$ is indexed $0, \ldots, l-1$.

2.2 Random Oracles and Official Generator Setup

The scalar random oracles H_{non} and H_{sig} are independent and return uniform elements of $\mathbb{F}_q$, whereas the authorization challenge oracles H_y, H_z, H_x return independent uniform elements of $\mathbb{F}_q^\times = \mathbb{F}_q \setminus \{0\}$. The roster digest H_{roster} is a domain-separated public hash. Every random-oracle query is lazy-sampled on first evaluation and cached thereafter. Distinct domain tags are never identified.

Definition 2.1 (Official proof-generator setup and d-generator DLR assumption)**.** For a public generator derivation context $(\mathsf{suite}, \sigma_{\text{gen}}, N)$, define the $d = 2N+1$ canonical official generator addresses

$$\mathfrak{a}_k = \mathsf{enc}(\mathsf{GEN}, \mathsf{suite}, \sigma_{\text{gen}}, N, k) \qquad (k \in [d]).$$

The protocol and security experiment expose group-valued generator queries exclusively on the fixed domain $\{\mathfrak{a}_1, \ldots, \mathfrak{a}_d\}$ (no auxiliary generator addresses outside $[d]$ exist in the interface). Designate the $d = 2N+1$ official proof generators as

$$(M_1, \ldots, M_d) = (H_0, G_1, \ldots, G_N, H_1, \ldots, H_N), \qquad M_k = H_{\text{gen}}(\mathfrak{a}_k) \in \mathbb{G} \setminus \{\mathcal{O}\}, \tag{3}$$

writing $\mathbf{G} = (G_1, \ldots, G_N)$ and $\mathbf{H} = (H_1, \ldots, H_N)$. We consider two alternative generator distributions:

(a) *Ideal random generator model:* Each $M_k = H_{\text{gen}}(\mathfrak{a}_k)$ $(k \in [d])$ is sampled independently and uniformly from $\mathbb{G} \setminus \{\mathcal{O}\}$ without disclosing $\log_G(M_k)$.

(b) *Fixed deterministic generator tuple (d-DLR model):* The tuple $(M_1, \ldots, M_d) \in (\mathbb{G} \setminus \{\mathcal{O}\})^d$ is derived deterministically via a public hash-to-curve procedure. For this fixed tuple, we define $\mathsf{Adv}_{d\text{-}\mathsf{DLR}}$ as the maximum probability that an efficient adversary outputs $(a, \eta_1, \ldots, \eta_d) \in \mathbb{F}_q^{1+d}$ with $(\eta_1, \ldots, \eta_d) \neq \mathbf{0}$ satisfying $aG + \sum_{k=1}^{d} \eta_k M_k = \mathcal{O}$ in $\mathbb{G}$. Under setup (a), Lemma 6.2 proves $\mathsf{Adv}_{d\text{-}\mathsf{DLR}} \leq d\,\mathsf{Adv}_{\mathsf{DLP}}$; under setup (b), hardness of d-DLR is an explicit assumption on the fixed generator tuple.

2.3 Representation-Annotated Soundness Model and Composite Propagation Rule

Definition 2.2 (First-evaluation annotated $\mathsf{AGM}+\mathsf{ROM}$ over the official public dictionary)**.** At any step of an execution, the public group dictionary $\mathcal{D}_{\text{pub}}$ is the ordered list of group elements supplied to the adversary's input tape thus far:

$$\mathcal{D}_{\text{pub}} = \Big(G,\ M_1, \ldots, M_d,\ (PK_i)_{i\in[n]},\ (U_{s,1}, U_{s,2})_{s \leq Q_1}\Big),$$

comprising the base generator G, the $d = 2N+1$ official generators $M_1, \ldots, M_d$, all registered public keys $(PK_i)_{i\in[n]}$, and all Round-1 nonce commitments $(U_{s,1}, U_{s,2})$ published by honest signers thus far. Every adversarially supplied group point $P \in \mathbb{G}$ in a protocol message (including Round-2 candidate aggregate points R_1, R_2), final forgery output, or direct scalar-oracle query must be accompanied on the adversary's annotation tape by a valid linear representation vector $\vec{\gamma}_P \in \mathbb{F}_q^{|\mathcal{D}_{\text{pub}}|}$ over the current dictionary $\mathcal{D}_{\text{pub}}$ satisfying $P = \langle \vec{\gamma}_P, \mathcal{D}_{\text{pub}} \rangle$ in $\mathbb{G}$ (no unaccounted auxiliary group atoms outside $\mathcal{D}_{\text{pub}}$ are permitted).

(i) *First-evaluation freezing and composite honest-signer propagation:* At the first evaluation of each canonical encoded scalar-oracle input string in, the experiment permanently

freezes the $\mathcal{D}_{\text{pub}}$-representations of all group arguments encoded in in (padding with zeros for any atoms appended to $\mathcal{D}_{\text{pub}}$ at later steps) prior to sampling and returning the random-oracle scalar. Whenever an honest signer executing Line 16 of Algorithm 1 triggers the first evaluation of $H_{\text{sig}}(\mathsf{enc}(\mathsf{CPK}, Q_e, t, R, m))$, the experiment derives and freezes the representation of $R = R_1 + \beta R_2$ by linear composition of the representations $\vec{\gamma}_{R_1}, \vec{\gamma}_{R_2}$ frozen at Line 12:

$$\vec{\gamma}_R := \vec{\gamma}_{R_1} + \beta \vec{\gamma}_{R_2} \pmod{q}.$$

Subsequent evaluations of an identical input string in return the cached scalar and preserve the frozen first-evaluation record.

(ii) *Syntactic opaque-atom rule:* Each group element M_k, honest public key PK_i ($i \in H$), and honest nonce commitment $U_{s,b}$ is a single formal basis symbol in $\mathcal{D}_{\text{pub}}$. When an external security reduction $\mathcal{B}$ internally synthesizes these group elements as linear combinations of an external challenge vector $\mathbf{W} = (W_1, \ldots, W_M) \in \mathbb{G}^M$, $\mathcal{B}$ writes only the resulting group elements $P \in \mathbb{G}$ onto $\mathcal{A}$'s input tape, never $\mathbf{W}$ or the internal coefficient matrices. Consequently, $\mathcal{A}$'s annotation vectors $\vec{\gamma}_P$ are syntactically defined over $\mathcal{D}_{\text{pub}}$ and never contain coordinates over $\mathbf{W}$.

3 Registration and Ordered Roster Assumption

Assumption 3.1 (Ideal extractable public-key registration with fixed ordering)**.** Setup and registration proceed in three sequential phases prior to any signing queries:

(1) *Static corruption declaration:* The adversary $\mathcal{A}$ declares a static corruption set $C \subseteq [n]$ with $|C| < t$. Let $H := [n] \setminus C$ be the set of honest participant indices.

(2) *Honest key generation:* For each $i \in H$ in increasing index order, the challenger samples $x_i \leftarrow \mathbb{F}_q^\times$ conditioned on $PK_i := x_i G \notin \{PK_j : j \in H, j < i\}$ (resampling x_i on collision) and publishes $(PK_i)_{i \in H}$.

(3) *Corrupted key registration:* For each $i \in C$, the adversary submits a candidate public key $PK_i \in \mathbb{G}$ together with an extracted scalar witness $x_i \in \mathbb{F}_q^\times$ satisfying $PK_i = x_i G$ to the registration functionality. The registry rejects the submission (and aborts setup if $\mathcal{A}$ fails to supply a valid roster) if $PK_i = \mathcal{O}$, $PK_i \neq x_i G$, or PK_i duplicates any honest key $\{PK_j : j \in H\}$ or previously registered corrupt key $\{PK_j : j \in C, j < i\}$.

Upon completion, the accepted roster $(PK_1, \ldots, PK_n) \in (\mathbb{G} \setminus \{\mathcal{O}\})^n$ consists of n pairwise distinct nonidentity points, with known exponents $(x_i)_{i \in C}$ provided to the reduction. No adaptive corruption is permitted.

In a practical deployment, Assumption 3.1 may be heuristically instantiated via a domain-separated Schnorr proof of possession (PoP):

$$B_i = r_i G, \qquad e_i = H_{\mathsf{PoP}}(\mathsf{enc}(\mathsf{PoP}, \mathsf{CPK}_{\text{ctx}}, i, PK_i, B_i)), \qquad z_i = r_i + e_i x_i \pmod{q}, \tag{4}$$

verified by $PK_i \neq \mathcal{O}$ and $z_i G = B_i + e_i PK_i$ [3]. Concrete PoP extraction is outside the scope of this manuscript; all theorems invoke Assumption 3.1 directly.

4 Protocol Construction

4.1 Roster, Masked Aggregate Key, and Session Context

Let $(PK_1, \ldots, PK_n) \in (\mathbb{G} \setminus \{\mathcal{O}\})^n$ be the ordered, duplicate-free registered roster produced under Assumption 3.1, and define the roster digest

$$\mathsf{CPK} = H_{\text{roster}}(\mathsf{enc}(n, PK_1, \ldots, PK_n)). \tag{5}$$

To coordinate a signature on message m for an authorized subset $S \subseteq [n]$ of cardinality $|S| \geq t$, the coordinator samples a mask $\xi \leftarrow \mathbb{F}_q$ uniformly and defines the masked aggregate verification key

$$Q_S = \sum_{i \in S} PK_i, \qquad Q_e = Q_S + \xi G, \tag{6}$$

resampling $\xi \leftarrow \mathbb{F}_q$ if $Q_e = \mathcal{O}$. For a signing session with unique session identifier sid, message m, candidate subset S, and valid nonidentity point $Q_e \neq \mathcal{O}$, the immutable session context is

$$\mathsf{ctx} = \mathsf{enc}\big(\mathsf{sid}, \mathsf{CPK}, t, m, Q_e, \mathsf{enc}(S)\big). \tag{7}$$

An adversarial coordinator may request interactive signing on any valid nonidentity point $Q_e \in \mathbb{G} \setminus \{\mathcal{O}\}$; honest signers do not verify checks (C1)–(C4) or check whether Q_e is linked to S during interactive signing. Threshold authorization is enforced publicly at final signature verification.

4.2 Two-Round Interactive Signing

Each honest participant $i \in H$ maintains the one-use two-nonce state machine specified in Algorithm 1.

Algorithm 1: Honest signer i: one-use two-nonce state machine

Round 1 input: $(\mathsf{sid}, S, m, Q_e)$.
if sid was previously used by signer i, $i \notin S$, $|S| < t$, the roster context is invalid, or $Q_e = \mathcal{O}$ **then**
    return abort.
end if
Mark sid permanently used at signer i; compute and store immutable ctx from (7).
Sample $(r_{i,1}, r_{i,2}) \leftarrow (\mathbb{F}_q^{\times})^2$ independently and store them as live one-use secret state for sid.
Return $U_{i,1} = r_{i,1}G$ and $U_{i,2} = r_{i,2}G$.
Round 2 input: Candidate aggregate nonce points (R_1, R_2) for session sid.
if session sid is not live, its context changed, or either R_1 or R_2 is invalid or equals $\mathcal{O}$ **then**
    Erase live nonce state $(r_{i,1}, r_{i,2})$; **return** abort.
end if
Compute $\beta \leftarrow H_{\mathrm{non}}(\mathsf{enc}(\mathsf{ctx}, R_1, R_2))$ and $R \leftarrow R_1 + \beta R_2$.
if $R = \mathcal{O}$ **then**
    Erase live nonce state $(r_{i,1}, r_{i,2})$; **return** abort.
end if
Compute $c \leftarrow H_{\mathrm{sig}}(\mathsf{enc}(\mathsf{CPK}, Q_e, t, R, m))$.
Compute $s_i \leftarrow r_{i,1} + \beta r_{i,2} + c x_i \pmod{q}$.
Erase $(r_{i,1}, r_{i,2})$ permanently, mark session sid replied, and **return** s_i.

For honest coordination on subset S, the coordinator collects $(U_{i,1}, U_{i,2})$ for $i \in S$, forms $R_1 = \sum_{i \in S} U_{i,1}$ and $R_2 = \sum_{i \in S} U_{i,2}$, and aborts if $R_1 = \mathcal{O}$, $R_2 = \mathcal{O}$, or $R = R_1 + \beta R_2 = \mathcal{O}$

where $\beta = H_{\text{non}}(\mathsf{enc}(\mathsf{ctx}, R_1, R_2))$. Otherwise, with

$$c = H_{\text{sig}}\big(\mathsf{enc}(\mathsf{CPK}, Q_e, t, R, m)\big), \tag{8}$$

the coordinator receives the partial responses $(s_i)_{i\in S}$ and computes the masked aggregate signature scalar

$$s = \sum_{i\in S} s_i \pmod q, \qquad s_e = s + c\xi \pmod q, \tag{9}$$

which satisfies the public Schnorr verification equation

$$s_e G = R + cQ_e. \tag{10}$$

4.3 Threshold-Authorization Witness and Noninteractive Prover

Define the public vectors $\chi, w \in \mathbb{F}_q^N$ and the padded public-key tuple $PK_{\text{pad}} \in \mathbb{G}^N$ by

$$\chi = (\underbrace{1, \ldots, 1}_{n+l}, 0) \in \mathbb{F}_q^N, \tag{11}$$

$$w = (\underbrace{1, \ldots, 1}_{n}, -1, -2, \ldots, -2^{l-1}, 0) \in \mathbb{F}_q^N, \tag{12}$$

$$PK_{\text{pad}} = (PK_1, \ldots, PK_n, \underbrace{\mathcal{O}, \ldots, \mathcal{O}}_{l}, G) \in \mathbb{G}^N. \tag{13}$$

A valid threshold-authorization witness is a tuple (b, d_s, ξ) with selector vector $b \in \{0,1\}^n$, binary surplus decomposition $d_s \in \{0,1\}^l$, and mask $\xi \in \mathbb{F}_q$ satisfying both the exact integer threshold identity

$$\sum_{i=1}^{n} b_i - t = \sum_{j=0}^{l-1} 2^j (d_s)_j \quad \text{in } \mathbb{Z} \tag{14}$$

and the exact group linkage identity

$$Q_e = \sum_{i=1}^{n} b_i PK_i + \xi G \quad \text{in } \mathbb{G}. \tag{15}$$

From (b, d_s, ξ), define the witness vectors $a_L, a_R \in \mathbb{F}_q^N$ by

$$a_L = (b \,\|\, d_s \,\|\, \xi), \qquad a_R = \big((b - 1_n) \,\|\, (d_s - 1_l) \,\|\, 0\big). \tag{16}$$

For the public authorization statement

$$\mathsf{st} = \mathsf{enc}(\mathsf{CPK}, n, t, m, Q_e), \tag{17}$$

the noninteractive prover $\mathsf{Auth.Prove}(\mathsf{st}; b, d_s, \xi)$ is specified in Algorithm 2.

Algorithm 2: Authorization prover $\mathsf{Auth.Prove}(\mathsf{st}; b, d_s, \xi)$

1: Sample blinding vectors $s_L, s_R \leftarrow \mathbb{F}_q^N$ and scalars $\alpha, \rho, \tau_1, \tau_2 \leftarrow \mathbb{F}_q$ independently.

2: Compute $A \leftarrow \alpha H_0 + \langle a_L, \mathbf{G}\rangle + \langle a_R, \mathbf{H}\rangle$.

3: Compute $S_{\mathrm{com}} \leftarrow \rho H_0 + \langle s_L, \mathbf{G}\rangle + \langle s_R, \mathbf{H}\rangle$.

4: Compute $Q_s \leftarrow \langle s_L, PK_{\mathrm{pad}}\rangle$ and set $T_0 \leftarrow (\mathsf{st}, A, S_{\mathrm{com}}, Q_s)$.

5: Query $y \leftarrow H_y(\mathsf{enc}(T_0))$ and $z \leftarrow H_z(\mathsf{enc}(T_0, y))$.

6: Set $\mathbf{y} \leftarrow (1, y, \ldots, y^{N-1}) \in \mathbb{F}_q^N$.

7: Define the linear vector polynomials $l(X) = a_L - z\chi + Xs_L$ and $r(X) = \mathbf{y} \circ (a_R + z\chi + Xs_R) + z^2 w$.

8: Expand the inner-product polynomial $T(X) = \langle l(X), r(X)\rangle = t_0 + t_1 X + t_2 X^2$ over $\mathbb{F}_q[X]$.

9: Compute $T_1 \leftarrow t_1 G + \tau_1 H_0$ and $T_2 \leftarrow t_2 G + \tau_2 H_0$.

10: Query $x \leftarrow H_x(\mathsf{enc}(T_0, y, z, T_1, T_2))$.

11: Compute $v_L \leftarrow a_L + xs_L$, $v_R \leftarrow a_R + xs_R$, and $\mu \leftarrow \alpha + x\rho \pmod q$.

12: Compute $\tau_x \leftarrow x\tau_1 + x^2\tau_2 \pmod q$ and $\hat{t} \leftarrow \langle v_L - z\chi,\ \mathbf{y} \circ (v_R + z\chi) + z^2 w\rangle \pmod q$.

13: **Return** $\pi_{\mathrm{Auth}} = (A, S_{\mathrm{com}}, Q_s, T_1, T_2, v_L, v_R, \hat{t}, \mu, \tau_x)$.

4.4 Authorization Verifier and Final Signature

Given st and π_{Auth}, the authorization verifier reconstructs $T_0 = (\mathsf{st}, A, S_{\mathrm{com}}, Q_s)$, queries $y = H_y(\mathsf{enc}(T_0))$, $z = H_z(\mathsf{enc}(T_0, y))$, $x = H_x(\mathsf{enc}(T_0, y, z, T_1, T_2))$, forms $\mathbf{y} = (1, y, \ldots, y^{N-1})$, and sets

$$\delta_\chi(y, z) = (z - z^2)\langle \chi, \mathbf{y}\rangle - z^3\langle \chi, w\rangle \pmod q, \tag{18}$$

$$\theta(y, z) = z^2 t + \delta_\chi(y, z) \pmod q. \tag{19}$$

The verifier accepts π_{Auth} if and only if all four checks (C1)–(C4) hold simultaneously:

$$A + xS_{\mathrm{com}} = \mu H_0 + \langle v_L, \mathbf{G}\rangle + \langle v_R, \mathbf{H}\rangle, \tag{C1}$$

$$Q_e + xQ_s = \langle v_L, PK_{\mathrm{pad}}\rangle, \tag{C2}$$

$$\hat{t} = \langle v_L - z\chi,\ \mathbf{y} \circ (v_R + z\chi) + z^2 w\rangle, \tag{C3}$$

$$\hat{t}G + \tau_x H_0 = \theta(y, z)G + xT_1 + x^2 T_2. \tag{C4}$$

The complete signature on message m is

$$\sigma = (Q_e, R, s_e, \pi_{\mathrm{Auth}}). \tag{20}$$

The public verification algorithm $\mathsf{Verify}(\mathsf{CPK}, n, t, m, \sigma)$ checks canonical encoding and group membership. It rejects $Q_e = \mathcal{O}$ or $R = \mathcal{O}$ and checks the Schnorr equation $s_e G = R + cQ_e$ (10), with $c = H_{\mathrm{sig}}(\mathsf{enc}(\mathsf{CPK}, Q_e, t, R, m))$, as well as all four authorization checks (C1)–(C4).

5 Algebraic Correctness

Proposition 5.1 (Honest completeness)**.** *For any valid authorization witness (b, d_s, ξ) and any honest, nonaborting interactive signing execution for the selected set $S = \{i \in [n] : b_i = 1\}$, $\mathsf{Verify}(\mathsf{CPK}, n, t, m, \sigma) = 1$.*

Proof. For the Schnorr equation (10), expanding s_e from (9) and Algorithm 1 gives

$$s_e G = \sum_{i\in S}(r_{i,1} + \beta r_{i,2} + c x_i)G + c\xi G = (R_1 + \beta R_2) + c\Big(\sum_{i\in S} PK_i + \xi G\Big) = R + cQ_e. \tag{21}$$

For (C1) and (C2), linearity of inner products gives $A + xS_{\text{com}} = (\alpha + x\rho)H_0 + \langle a_L + xs_L, \mathbf{G}\rangle + \langle a_R + xs_R, \mathbf{H}\rangle$ and $Q_e + xQ_s = \langle a_L + xs_L, PK_{\text{pad}}\rangle$. Step 12 of Algorithm 2 ensures (C3). For (C4), expanding the constant term $t_0 = T(0) = \langle a_L - z\chi,\ \mathbf{y} \circ (a_R + z\chi) + z^2 w\rangle$ yields

$$t_0 = \langle a_L \circ a_R, \mathbf{y}\rangle + z\langle \chi \circ (a_L - a_R), \mathbf{y}\rangle - z^2\langle \chi, \mathbf{y}\rangle + z^2\langle a_L, w\rangle - z^3\langle \chi, w\rangle. \tag{22}$$

Since $a_L \circ a_R = \mathbf{0}$, $\chi \circ (a_L - a_R) = \chi$, and $\langle a_L, w\rangle = t$, equation (22) simplifies to $t_0 = z^2 t + \delta_\chi(y,z) = \theta(y,z)$. Because $\hat{t} = T(x) = t_0 + xt_1 + x^2 t_2$ and $\tau_x = x\tau_1 + x^2\tau_2$, we have $\hat{t}G + \tau_x H_0 = \theta(y,z)G + x(t_1 G + \tau_1 H_0) + x^2(t_2 G + \tau_2 H_0) = \theta(y,z)G + xT_1 + x^2 T_2$, verifying (C4). □

6 Threshold Knowledge Soundness and Exact Public-Key Linkage

6.1 The Security Objective and Generator-Relation Event

Definition 6.1 (Authorization knowledge soundness with linkage)**.** The adversary supplies a statement $\mathsf{st} = \mathsf{enc}(\mathsf{CPK}, n, t, m, Q_e)$ and a certificate $\pi_{\text{Auth}} = (A, S_{\text{com}}, Q_s, T_1, T_2, v_L, v_R, \hat{t}, \mu, \tau_x)$ accepted by checks (C1)–(C4). In the annotated experiment of Definition 2.2, a straight-line extractor reads the frozen first-evaluation representation of (A, S_{com}) at the first evaluation of $H_y(\mathsf{enc}(T_0))$ and outputs (b, d_s, ξ). Extraction succeeds—denoted $\neg\mathsf{Fail}_{\text{ext}}$—if and only if (b, d_s, ξ) satisfies both the integer threshold relation (14) and the exact group linkage equality (15).

Let Π_M denote the formal coordinate projection onto the $d = 2N + 1$ official proof generators $(M_1, \ldots, M_d) = (H_0, G_1, \ldots, G_N, H_1, \ldots, H_N)$. For a completed challenge chain, write $T_0 = (\mathsf{st}, A, S_{\text{com}}, Q_s)$. At the first evaluation of $H_y(\mathsf{enc}(T_0))$, the experiment freezes the $\mathcal{D}_{\text{pub}}$-representations of $A, S_{\text{com}}, Q_e, Q_s$; similarly, at the first evaluation of $H_x(\mathsf{enc}(T_0, y, z, T_1, T_2))$, it freezes those of T_1, T_2. More generally, when an input to H_z or H_x contains T_0 before the corresponding H_y call, the first-appearance representations are recorded, but that early order is accounted for by $\mathsf{Bad}_{y\to z}$ or $\mathsf{Bad}_{z\to x}$ if the chain subsequently completes with matching challenges. Therefore, on every completed chain outside those bad-order events, $H_y(\mathsf{enc}(T_0))$ precedes the matching H_x first evaluation, and the extracted a_L, s_L and the elements D_0, D_1 of Lemma 6.4 are fixed before its fresh challenge x is sampled. No unconditional chronology is asserted for chains discarded as out-of-order. Write these frozen representations as

$$\begin{aligned} A &= \alpha H_0 + \langle a_L, \mathbf{G}\rangle + \langle a_R, \mathbf{H}\rangle + E_A, & S_{\text{com}} &= \rho H_0 + \langle s_L, \mathbf{G}\rangle + \langle s_R, \mathbf{H}\rangle + E_S, \\ T_1 &= \tau_1 H_0 + \langle u_{1,L}, \mathbf{G}\rangle + \langle u_{1,R}, \mathbf{H}\rangle + E_{T_1}, & T_2 &= \tau_2 H_0 + \langle u_{2,L}, \mathbf{G}\rangle + \langle u_{2,R}, \mathbf{H}\rangle + E_{T_2}, \end{aligned} \tag{24}$$

where $\Pi_M(E_A) = \Pi_M(E_S) = \Pi_M(E_{T_1}) = \Pi_M(E_{T_2}) = \mathbf{0}$. When an accepted certificate satisfies (C1) and (C4), the two group differences

$$\begin{aligned} \Delta_{\text{C1}} &:= \mu H_0 + \langle v_L, \mathbf{G}\rangle + \langle v_R, \mathbf{H}\rangle - A - xS_{\text{com}} = \mathcal{O}, \\ \Delta_{\text{C4}} &:= (\hat{t} - \theta(y,z))G + \tau_x H_0 - xT_1 - x^2 T_2 = \mathcal{O}. \end{aligned}$$

are verified equalities to $\mathcal{O}$ in $\mathbb{G}$. Substituting the frozen representations (24) into $\Delta_{\text{C1}} = \mathcal{O}$ and $\Delta_{\text{C4}} = \mathcal{O}$ expresses each as a linear combination of $\mathcal{D}_{\text{pub}}$ equal to $\mathcal{O}$. Define the

checkable authorization generator-relation event G_{auth} to hold on an accepted certificate if either $\Pi_M(\Delta_{\text{C1}}) \neq \mathbf{0}$ or $\Pi_M(\Delta_{\text{C4}}) \neq \mathbf{0}$, where

$$\Pi_M(\Delta_{\text{C1}}) = (\mu - \alpha - x\rho,\ v_L - a_L - xs_L,\ v_R - a_R - xs_R),$$
$$\Pi_M(\Delta_{\text{C4}}) = (\tau_x - x\tau_1 - x^2\tau_2,\ -xu_{1,L} - x^2u_{2,L},\ -xu_{1,R} - x^2u_{2,R}).$$

In any reduction that knows the base-G discrete logarithms of all non-M_k public atoms appearing in $E_A, E_S, E_{T_1}, E_{T_2}$ (which by Definition 2.2 consist exclusively of G, $(PK_i)_{i\in[n]}$, and $(U_{s,1}, U_{s,2})_{s\le Q_1}$), the occurrence of G_{auth} on an accepted certificate directly solves the d-DLR problem of Definition 2.1:

$$aG + \sum_{k=1}^{d} \eta_k M_k = \mathcal{O}, \qquad (\eta_1, \ldots, \eta_d) \neq \mathbf{0}. \tag{23}$$

Lemma 6.2 (Reduction of d-DLR to DLP for random generators)**.** *When the official generators $(M_1, \ldots, M_d) \leftarrow (\mathbb{G} \setminus \{\mathcal{O}\})^d$ are sampled uniformly and independently at setup (Definition 2.1), any solver finding a relation (23) with probability $\mathsf{Adv}_{d\text{-DLR}}$ yields a DLP solver in $\mathbb{G}$ with success probability at least $\mathsf{Adv}_{d\text{-DLR}}/d$; that is, $\mathsf{Adv}_{d\text{-DLR}} \le d\,\mathsf{Adv}_{\mathsf{DLP}}$.*

Proof. Given a DLP instance $Y = uG \in \mathbb{G} \setminus \{\mathcal{O}\}$, the reduction samples a uniform index $k^* \leftarrow [d]$, sets $M_{k^*} := Y$, and for all $k \in [d] \setminus \{k^*\}$ samples $\gamma_k \leftarrow \mathbb{F}_q^\times$ and sets $M_k := \gamma_k G$. The joint distribution of $(M_1, \ldots, M_d)$ is uniform over $(\mathbb{G}\setminus\{\mathcal{O}\})^d$, and k^* is information-theoretically hidden from the solver. Whenever the solver outputs a relation (23) with $(\eta_1, \ldots, \eta_d) \neq \mathbf{0}$, the hidden index k^* falls in the nonzero support $\{\, k \in [d] : \eta_k \neq 0 \,\}$ with conditional probability at least $1/d$, allowing the reduction to compute $u = -\eta_{k^*}^{-1}(a + \sum_{k\neq k^*} \eta_k\gamma_k) \pmod q$. □

6.2 Frozen-Coordinate Extraction and Ordered Oracle Events

Lemma 6.3 (Commitment-coordinate consistency)**.** *On any accepted certificate outside G_{auth}, the frozen coordinates in (24) satisfy*

$$v_L = a_L + xs_L, \qquad v_R = a_R + xs_R, \qquad \mu = \alpha + x\rho, \qquad \tau_x = x\tau_1 + x^2\tau_2 \pmod q. \tag{25}$$

Proof. Outside G_{auth}, we have $\Pi_M(\Delta_{\text{C1}}) = \mathbf{0}$ and $\Pi_M(\Delta_{\text{C4}}) = \mathbf{0}$ by definition, whose H_0, $\mathbf{G}$, and $\mathbf{H}$ components immediately give (25). □

To analyze the random-oracle challenges $y = H_y(\mathsf{enc}(T_0))$, $z = H_z(\mathsf{enc}(T_0, y))$, and $x = H_x(\mathsf{enc}(T_0, y, z, T_1, T_2))$ across Q_y, Q_z, Q_x distinct first evaluations, we define two explicit premature-query (out-of-order) events:

- $\mathsf{Bad}_{y\to z}$: At some first evaluation of H_y on an input $\mathsf{enc}(T_0)$ returning a fresh scalar $y \leftarrow \mathbb{F}_q^\times$, the pair $\mathsf{enc}(T_0, y)$ had already been evaluated at H_z. Because at most Q_z prior H_z inputs exist and injectivity of enc fixes at most one candidate $y \in \mathbb{F}_q^\times$ per prior H_z query, a union bound over all Q_y first evaluations gives $\Pr[\mathsf{Bad}_{y\to z}] \le \frac{Q_yQ_z}{q-1}$.
- $\mathsf{Bad}_{z\to x}$: At some first evaluation of H_z on an input $\mathsf{enc}(T_0, y)$ returning a fresh scalar $z \leftarrow \mathbb{F}_q^\times$, an input of the form $\mathsf{enc}(T_0, y, z, T_1, T_2)$ had already been evaluated at H_x. Because at most Q_x prior H_x inputs exist and injectivity of enc fixes at most one candidate $z \in \mathbb{F}_q^\times$ per prior H_x query, a union bound over all Q_z first evaluations gives $\Pr[\mathsf{Bad}_{z\to x}] \le \frac{Q_zQ_x}{q-1}$.

For any certificate accepted by Verify, the verifier evaluates the exact chain $y = H_y(\mathsf{enc}(T_0))$, $z = H_z(\mathsf{enc}(T_0, y))$, and $x = H_x(\mathsf{enc}(T_0, y, z, T_1, T_2))$. Let t_y, t_z, t_x be the first-evaluation times of these three exact inputs, including evaluations during verification. If $t_z < t_y$, the fresh output y matched a prior H_z input and triggered $\mathsf{Bad}_{y\to z}$; if $t_x < t_z$, the fresh output z matched a prior H_x input and triggered $\mathsf{Bad}_{z\to x}$. Thus, outside their union, any accepted certificate has $t_y < t_z < t_x$. In particular, the frozen representations of T_0 precede the fresh x draw used in Lemma 6.4.

Lemma 6.4 (Exact key linkage)**.** *Outside $G_{auth} \cup \mathsf{Bad}_{y\to z} \cup \mathsf{Bad}_{z\to x}$, acceptance of (C2) implies*

$$Q_e = \langle a_L, PK_{pad} \rangle \quad \text{in } \mathbb{G}, \tag{26}$$

except on a first-query root event Bad_{link} over the Q_x first evaluations of H_x of probability at most $\Pr[\mathsf{Bad}_{link}] \le \frac{Q_x}{q-1}$.

Proof. For each first evaluation of H_x on an input $\mathsf{enc}(T_0, y, z, T_1, T_2)$ whose $T_0 = (\mathsf{st}, A, S_{\mathrm{com}}, Q_s)$ has already been evaluated at $H_y(\mathsf{enc}(T_0))$, the group elements $D_0 := Q_e - \langle a_L, PK_{\mathrm{pad}} \rangle \in \mathbb{G}$ and $D_1 := Q_s - \langle s_L, PK_{\mathrm{pad}} \rangle \in \mathbb{G}$ are completely determined prior to sampling the fresh scalar $x \leftarrow \mathbb{F}_q^\times$. If $D_0 \neq \mathcal{O}$, there is at most one scalar $x \in \mathbb{F}_q^\times$ satisfying $D_0 + xD_1 = \mathcal{O}$ in the prime-order group $\mathbb{G}$; let $\mathsf{Bad}_{\mathrm{link}}$ be the event that the fresh draw $x \leftarrow \mathbb{F}_q^\times$ hits this root on any of the Q_x first evaluations of H_x. By a union bound, $\Pr[\mathsf{Bad}_{\mathrm{link}}] \le \frac{Q_x}{q-1}$. On any accepted certificate outside $G_{\mathrm{auth}} \cup \mathsf{Bad}_{y\to z} \cup \mathsf{Bad}_{z\to x} \cup \mathsf{Bad}_{\mathrm{link}}$, substituting $v_L = a_L + xs_L$ from (25) into (C2) ($Q_e + xQ_s = \langle v_L, PK_{\mathrm{pad}} \rangle$) gives $D_0 + xD_1 = \mathcal{O}$, which forces $D_0 = \mathcal{O}$ and hence (26). □

6.3 The Threshold Polynomial

Lemma 6.5 (Polynomial constraint extraction)**.** *Under the annotated model of Definition 2.2, on any accepted certificate outside G_{auth}, the frozen vectors $a_L, a_R \in \mathbb{F}_q^N$ from (24) satisfy*

$$a_{L,i} a_{R,i} = 0, \qquad a_{L,i} - a_{R,i} = 1 \quad (1 \le i \le n + l), \qquad \langle a_L, w \rangle = t \pmod q, \tag{27}$$

except on an unconditional first-query oracle event Bad_{FS} (including $\mathsf{Bad}_{y\to z} \cup \mathsf{Bad}_{z\to x}$) of probability at most

$$\epsilon_{FS} := \Pr[\mathsf{Bad}_{FS}] \le \frac{(N-1)Q_y + 2Q_z + 2Q_x + Q_yQ_z + Q_zQ_x}{q-1}. \tag{28}$$

Proof. For each first evaluation of H_y on $\mathsf{enc}(T_0)$, the frozen vectors $a_L, a_R \in \mathbb{F}_q^N$ from (24) are fixed prior to sampling $y \leftarrow \mathbb{F}_q^\times$. Define the bivariate polynomial over $\mathbb{F}_q$:

$$F(Y, Z) := \langle a_L \circ a_R, \mathbf{Y} \rangle + Z\langle \chi \circ (a_L - a_R - 1_N), \mathbf{Y} \rangle + Z^2(\langle a_L, w \rangle - t), \qquad \mathbf{Y} := (1, Y, \ldots, Y^{N-1}). \tag{29}$$

We define three root events at the first evaluations of H_y, H_z, H_x:

1. Bad_y: At a first evaluation of $H_y(\mathsf{enc}(T_0))$, either $a_L \circ a_R \neq \mathbf{0}$ or $\chi \circ (a_L - a_R - 1_N) \neq \mathbf{0}$, and the fresh scalar $y \leftarrow \mathbb{F}_q^\times$ is a root of the first nonzero polynomial among $\langle a_L \circ a_R, \mathbf{Y} \rangle$ and $\langle \chi \circ (a_L - a_R - 1_N), \mathbf{Y} \rangle$. Since each has degree at most $N - 1$, a union bound over Q_y queries gives $\Pr[\mathsf{Bad}_y] \le \frac{(N-1)Q_y}{q-1}$.

2. Bad_z: At a first evaluation of $H_z(\mathsf{enc}(T_0, y))$ (where T_0 was already evaluated at H_y), the univariate polynomial $Z \mapsto F(y, Z)$ of degree at most 2 is not identically zero, and the fresh scalar $z \leftarrow \mathbb{F}_q^\times$ satisfies $F(y, z) = 0$. A union bound over Q_z queries gives $\Pr[\mathsf{Bad}_z] \le \frac{2Q_z}{q-1}$.

3. Bad_x: At a first evaluation of $H_x(\mathsf{enc}(T_0, y, z, T_1, T_2))$ (where T_0 and (T_0, y) were already evaluated at H_y and H_z), write $T_{\mathrm{ext}}(X) := \langle a_L - z\chi + Xs_L,\ \mathbf{y} \circ (a_R + z\chi + Xs_R) + z^2 w \rangle = t_0 + t_1 X + t_2 X^2$, where $t_0, t_1, t_2 \in \mathbb{F}_q$ and $\tau_1, \tau_2 \in \mathbb{F}_q$ from (24) are completely fixed prior to drawing $x \leftarrow \mathbb{F}_q^\times$. Consequently, the three group elements

   $$P_0 := (t_0 - \theta(y, z))G \in \mathbb{G}, \qquad P_1 := t_1 G + \tau_1 H_0 - T_1 \in \mathbb{G}, \qquad P_2 := t_2 G + \tau_2 H_0 - T_2 \in \mathbb{G}$$

   are fixed prior to sampling $x \leftarrow \mathbb{F}_q^\times$. We say Bad_x occurs if $t_0 \neq \theta(y, z) \pmod q$ (so $P_0 \neq \mathcal{O}$) and the fresh draw $x \leftarrow \mathbb{F}_q^\times$ satisfies $P_0 + xP_1 + x^2 P_2 = \mathcal{O}$ in $\mathbb{G}$. Because $\mathbb{G} = \langle G \rangle$ has prime order q, $P_0 + xP_1 + x^2 P_2 = \mathcal{O}$ is a quadratic equation $\log_G(P_0) + x \log_G(P_1) + x^2 \log_G(P_2) \equiv 0 \pmod q$ with nonzero constant term $\log_G(P_0) = t_0 - \theta(y, z) \neq 0$, which has at most 2 roots in $\mathbb{F}_q^\times$. A union bound over Q_x queries gives $\Pr[\mathsf{Bad}_x] \le \frac{2Q_x}{q-1}$.

Set $\mathsf{Bad}_{\mathrm{FS}} := \mathsf{Bad}_{y\to z} \cup \mathsf{Bad}_{z\to x} \cup \mathsf{Bad}_y \cup \mathsf{Bad}_z \cup \mathsf{Bad}_x$, which satisfies (28). On any accepted certificate outside $G_{\mathrm{auth}} \cup \mathsf{Bad}_{\mathrm{FS}}$, Lemma 6.3 and (C3) imply $\hat{t} = T_{\mathrm{ext}}(x) = t_0 + t_1 x + t_2 x^2$ and $\tau_x = x\tau_1 + x^2\tau_2$; substituting both into (C4) yields $P_0 + xP_1 + x^2P_2 = \mathcal{O}$. Outside Bad_x, this forces $t_0 = \theta(y,z) \pmod q$, which is equivalent to $F(y,z) = 0 \pmod q$. Outside Bad_z and Bad_y, the polynomial $F(Y,Z)$ is identically zero, so all its coefficients in (29) vanish. Because $\chi_i = 1$ for $1 \le i \le n+l$, this yields (27). □

Lemma 6.6 (Field-to-integer threshold lifting)**.** *Assume the parameter bound (2). If $a_L, a_R \in \mathbb{F}_q^N$ satisfy (27), then writing $b = (a_{L,1}, \dots, a_{L,n})$, $d_s = (a_{L,n+1}, \dots, a_{L,n+l})$, and $\xi = a_{L,N}$ yields $b \in \{0,1\}^n$, $d_s \in \{0,1\}^l$, and $\sum_{i=1}^n b_i - t = \sum_{j=0}^{l-1} 2^j (d_s)_j$ over $\mathbb{Z}$; in particular, $\sum_{i=1}^n b_i \ge t$.*

Proof. For each $1 \le i \le n+l$, substituting $a_{R,i} = a_{L,i} - 1$ into $a_{L,i}a_{R,i} = 0$ gives $a_{L,i}(a_{L,i}-1) = 0$ in $\mathbb{F}_q$, so $a_{L,i} \in \{0,1\}$. By (12), $\langle a_L, w\rangle = t \pmod q$ gives $D := \sum_{i=1}^n b_i - t - \sum_{j=0}^{l-1} 2^j (d_s)_j \equiv 0 \pmod q$. Viewed as an integer, $-(n + 2^l - 1) \le D \le n - 1$. Under $q > n + 2^l - 1$ from (2), the only multiple of q in this interval is 0, so $D = 0$ in $\mathbb{Z}$ and $\sum_{i=1}^n b_i = t + \sum_{j=0}^{l-1} 2^j (d_s)_j \ge t$. □

Theorem 6.7 (Standalone authorization knowledge soundness and exact linkage)**.** *Assume the official-generator model of Definition 2.1, the first-evaluation annotated model of Definition 2.2, the ideal ordered extractable-registration interface of Assumption 3.1, and the parameter bound (2). Consider the* standalone authorization experiment *in which the challenger generates honest registration keys with known exponents, processes corrupted registrations using the extracted exponents, supplies honest nonce commitments with known exponents if signing queries are allowed, and finally checks one adversarial authorization certificate by (C1)–(C4). Write $\mathsf{Acc}_{\mathrm{auth}}$ for the event that all four checks accept.*

Across at most Q_y, Q_z, Q_x distinct first evaluations of H_y, H_z, H_x, respectively, let $\mathsf{Bad}_{\mathrm{auth}} := \mathsf{Bad}_{\mathrm{FS}} \cup \mathsf{Bad}_{\mathrm{link}}$. Then

$$\epsilon_{\mathrm{auth}} := \Pr[\mathsf{Bad}_{\mathrm{auth}}] \le \frac{(N-1)Q_y + 2Q_z + 3Q_x + Q_yQ_z + Q_zQ_x}{q-1}. \tag{30}$$

For every accepted (C1)–(C4) certificate outside $G_{\mathrm{auth}} \cup \mathsf{Bad}_{\mathrm{auth}}$, the frozen-coordinate straight-line extractor returns (b, d_s, ξ) satisfying

$$\begin{aligned} &b \in \{0,1\}^n, \qquad d_s \in \{0,1\}^l, \\ &\sum_{i=1}^n b_i - t = \sum_{j=0}^{l-1} 2^j (d_s)_j \quad \text{in } \mathbb{Z}, \\ &Q_e = \sum_{i=1}^n b_i PK_i + \xi G \quad \text{in } \mathbb{G}. \end{aligned}$$

In the standalone experiment, the generator-relation event is bounded by $\Pr[G_{\mathrm{auth}}] \le \mathsf{Adv}_{d\text{-}\mathsf{DLR}}$. Consequently,

$$\begin{aligned} &\Pr[\mathsf{Acc}_{\mathrm{auth}} \ \wedge \ \mathsf{Fail}_{\mathrm{ext}}] \\ &\qquad \le \epsilon_{\mathrm{auth}} + \mathsf{Adv}_{d\text{-}\mathsf{DLR}}. \end{aligned} \tag{30a}$$

For ideal independently random official generators, Lemma 6.2 further gives the upper bound $\epsilon_{\mathrm{auth}} + d\,\mathsf{Adv}_{\mathsf{DLP}}$; for a fixed deterministic generator tuple, d-DLR is retained as a separate hardness assumption.

Proof. By Lemmas 6.3–6.6, an accepted certificate outside G_{auth} and $\mathsf{Bad}_{\mathrm{auth}}$ extracts Boolean selection and surplus vectors satisfying the integer threshold identity. Equation (26), together with $PK_{\mathrm{pad}} = (PK_1, \dots, PK_n, \mathcal{O}, \dots, \mathcal{O}, G)$, yields $Q_e = \sum_{i=1}^n b_i PK_i + \xi G$. Thus

$$\{\mathsf{Acc}_{\mathrm{auth}}\} \cap \mathsf{Fail}_{\mathrm{ext}} \subseteq G_{\mathrm{auth}} \cup \mathsf{Bad}_{\mathrm{auth}}.$$

The first-query and linkage estimates in Lemmas 6.5 and 6.4 give (30).

For the standalone generator-relation term, construct a d-DLR solver $\mathcal{B}_{\text{auth-DLR}}$ given $(G, M_1, \ldots, M_d)$. It publishes those challenge generators as the entire official proof-generator tuple. It simulates the standalone experiment exactly: after the static corruption set is declared, it samples the honest exponents $x_i \in \mathbb{F}_q^\times$ with the roster's specified collision rejection, registers each corrupted key together with its ideal extracted exponent x_i, and, if honest signing queries exist, answers them using independently sampled one-use nonce exponents. In particular, it knows the base-G logarithms of every public dictionary atom *except* the official generators M_k. It runs the scalar random oracles by lazy sampling and maintains the mandated frozen first-evaluation representation records.

Suppose the adversary outputs a certificate accepted by (C1)–(C4) for which G_{auth} occurs. By definition of that event, at least one verified zero-group difference Δ_{C1} or Δ_{C4} has a nonzero formal coefficient vector under Π_M. The solver chooses the first such difference, substitutes the known logarithms of every nonofficial atom in its frozen representation, and obtains explicit $(a, \eta_1, \ldots, \eta_d)$ such that

$$aG + \sum_{k=1}^{d} \eta_k M_k = \mathcal{O}, \qquad (\eta_1, \ldots, \eta_d) \neq \mathbf{0}.$$

This is a valid d-DLR solution. Since the simulation matches the standalone authorization experiment, including key-validation and oracle ordering, its success probability is at least $\Pr[G_{\text{auth}}]$, which proves $\Pr[G_{\text{auth}}] \leq \mathsf{Adv}_{d\text{-DLR}}$. Taking a union bound with (30) establishes (30a); Lemma 6.2 gives the optional DLP substitution only under the ideal random-generator setup. □

7 Passive External-Verifier Signer-Set Privacy

7.1 Observer Model and Game

Definition 7.1 (Passive external-verifier signer-set privacy)**.** Fix an honestly registered public roster, threshold t, message m, and independent random oracles. A passive external-verifier adversary $\mathcal{A}$ chooses two subsets $S_0, S_1 \subseteq [n]$ with $|S_0| \geq t$ and $|S_1| \geq t$. The challenger samples $b \leftarrow \{0, 1\}$, honestly executes one nonaborting signing and authorization run for subset S_b using a fresh mask $\xi \leftarrow \mathbb{F}_q$ conditioned on $Q_e \neq \mathcal{O}$, fresh one-use nonces conditioned on $R_1, R_2, R \neq \mathcal{O}$, and fresh authorization blinding randomness, and gives $\mathcal{A}$ only the final signature $\sigma_b = (Q_e, R, s_e, \pi_{\text{Auth}})$. The adversary knows the public roster, parameters, and message m and may query the public random oracles, but observes neither the coordinator's internal state, interactive session messages, individual nonce commitments, nor network metadata. Its distinguishing advantage is $\mathsf{Adv}_{\text{priv}}(\mathcal{A}) = |\Pr[b' = b] - \frac{1}{2}|$.

Proposition 7.2 (Perfect subset-independent masked-key distribution)**.** *For any fixed qualified subset $S \subseteq [n]$, if $\xi \leftarrow \mathbb{F}_q$ is resampled until $Q_e = \sum_{i \in S} PK_i + \xi G \neq \mathcal{O}$, then Q_e is uniformly distributed over $\mathbb{G} \setminus \{\mathcal{O}\}$, independently of S. Equivalent to this distribution is sampling $u \leftarrow \mathbb{F}_q^\times$ and setting $Q_e := uG$.*

Proof. In the prime-order group $\mathbb{G}$, the map $\xi \mapsto \sum_{i \in S} PK_i + \xi G$ is a bijection from $\mathbb{F}_q$ onto $\mathbb{G}$. Conditioning on $Q_e \neq \mathcal{O}$ removes the unique value of ξ mapping to $\mathcal{O}$, leaving the uniform distribution on $\mathbb{G} \setminus \{\mathcal{O}\}$, which is identically induced by $u \leftarrow \mathbb{F}_q^\times \mapsto uG$. □

7.2 Authorization Proof Simulation

Lemma 7.3 (Witness-free simulator for (C1)–(C4))**.** *In the programmable-ROM experiment where H_y, H_z, H_x are programmable at previously unqueried inputs, for every public statement* st *admitting a valid authorization witness there is a PPT simulator requiring no witness that outputs an accepting proof π_{Auth} whose distribution (conditioned on no prior query conflict) is*

identical to that of Auth.Prove. *If the passive distinguisher makes at most* $Q_y^{\mathcal{A}}, Q_z^{\mathcal{A}}, Q_x^{\mathcal{A}}$ *prior queries to* H_y, H_z, H_x*, the programming conflict probability is at most*

$$\epsilon_{prog} \leq \frac{Q_y^{\mathcal{A}} + Q_z^{\mathcal{A}} + Q_x^{\mathcal{A}}}{q}. \tag{31}$$

Proof. The simulator independently samples $y, z, x \leftarrow \mathbb{F}_q^\times$, $\tilde{v}_L, \tilde{v}_R \leftarrow \mathbb{F}_q^N$, $A, T_2 \leftarrow \mathbb{G}$, and $\mu, \tau_x \leftarrow \mathbb{F}_q$, sets $\mathbf{y} = (1, y, \ldots, y^{N-1})$, computes $\theta(y, z)$ via (19), and solves checks (C2), (C1), (C3), and (C4) algebraically via:

$$Q_s = x^{-1}(\langle \tilde{v}_L, PK_{\text{pad}} \rangle - Q_e), \tag{32}$$

$$S_{\text{com}} = x^{-1}(\mu H_0 + \langle \tilde{v}_L, \mathbf{G} \rangle + \langle \tilde{v}_R, \mathbf{H} \rangle - A), \tag{33}$$

$$\hat{t} = \langle \tilde{v}_L - z\chi,\ \mathbf{y} \circ (\tilde{v}_R + z\chi) + z^2 w \rangle, \tag{34}$$

$$T_1 = x^{-1}((\hat{t} - \theta(y, z))G + \tau_x H_0 - x^2 T_2). \tag{35}$$

Observe that (32) ensures $Q_e + xQ_s = \langle \tilde{v}_L, PK_{\text{pad}} \rangle$, satisfying (C2) identically. The simulator sets $T_0 = (\mathsf{st}, A, S_{\text{com}}, Q_s)$, programs $H_y(\mathsf{enc}(T_0)) = y$, $H_z(\mathsf{enc}(T_0, y)) = z$, and $H_x(\mathsf{enc}(T_0, y, z, T_1, T_2)) = x$, and outputs $\pi_{\text{Auth}} = (A, S_{\text{com}}, Q_s, T_1, T_2, \tilde{v}_L, \tilde{v}_R, \hat{t}, \mu, \tau_x)$. For any fixed valid witness and fixed challenges $(y, z, x) \in (\mathbb{F}_q^\times)^3$, the honest prover's blinding tuple $(s_L, s_R, \alpha, \rho, \tau_2, \tau_1) \in \mathbb{F}_q^{2N+4}$ maps bijectively to $(v_L, v_R, \log_G A, \mu, \log_G T_2, \tau_x)$ with Jacobian determinant $x^{2N}(\log_G H_0)^2 x^2 \neq 0$, while $(Q_s, S_{\text{com}}, \hat{t}, T_1)$ are uniquely determined by (C1)–(C4). Because A is uniform in $\mathbb{G}$ and appears in T_0, a union bound over prior oracle queries gives (31). □

7.3 The Honest Signing Part of the Passive View and Efficient Hybrid Simulation

Lemma 7.4 (Hidden aggregate-nonce pair)**.** *In an honest nonaborting signing session where* $\mathsf{enc}(\mathsf{ctx}, R_1, R_2)$ *is fresh when evaluated at* H_{non}*, conditioned on* $R_1 \neq \mathcal{O}$ *and* $R_2 \neq \mathcal{O}$*, the aggregate points* (R_1, R_2) *are independent uniform elements of* $\mathbb{G} \setminus \{\mathcal{O}\}$*, and conditioned further on* $R = R_1 + \beta R_2 \neq \mathcal{O}$ *(for* $\beta \leftarrow \mathbb{F}_q$*),* R *is uniform in* $\mathbb{G} \setminus \{\mathcal{O}\}$ *independently of* S*.*

Proof. Each $R_b = \sum_{i \in S} r_{i,b} G$ ($b \in \{1, 2\}$) is a sum of $|S| \geq 1$ independent uniform elements of $\mathbb{G} \setminus \{\mathcal{O}\}$. Multiplication by any nonzero scalar $\lambda \in \mathbb{F}_q^\times$ is a bijection on $\mathbb{G} \setminus \{\mathcal{O}\}$ that preserves the distribution of R_b; since $\mathbb{F}_q^\times$ acts transitively on $\mathbb{G} \setminus \{\mathcal{O}\}$, R_b is uniform on $\mathbb{G} \setminus \{\mathcal{O}\}$ when conditioned on $R_b \neq \mathcal{O}$ (which fails with probability at most $1/(q-1)$). Since $(r_{i,1})_{i \in S}$ and $(r_{i,2})_{i \in S}$ are sampled independently, R_1 and R_2 are independent. Finally, for any fixed $R_1, R_2 \in \mathbb{G} \setminus \{\mathcal{O}\}$, the map $\beta \mapsto R_1 + \beta R_2$ is a bijection from $\mathbb{F}_q$ onto $\mathbb{G}$, so a fresh $\beta \leftarrow \mathbb{F}_q$ makes R uniform on $\mathbb{G}$, and conditioning on $R \neq \mathcal{O}$ makes R uniform on $\mathbb{G} \setminus \{\mathcal{O}\}$, independently of S. □

Theorem 7.5 (Passive final-signature signer-subset privacy)**.** *Under the hypotheses of Definition 7.1, Lemma 7.3, and Lemma 7.4, if the passive external distinguisher* $\mathcal{A}$ *makes at most* $Q_{non}^{\mathcal{A}}$ *queries to* H_{non}*, its advantage satisfies*

$$\mathsf{Adv}_{priv}(\mathcal{A}) \leq 2\epsilon_{prog} + \frac{2Q_{non}^{\mathcal{A}}}{q-2} + \frac{6}{q-1}. \tag{36}$$

Proof. For a fixed bit $b \in \{0, 1\}$, we define a sequence of efficient polynomial-time hybrids:

- $\mathsf{Hyb}_0^{(b)}$: The real privacy experiment of Definition 7.1 for subset S_b.
- $\mathsf{Hyb}_1^{(b)}$: Replace $\mathsf{Auth.Prove}(\mathsf{st}; b_{S_b}, d_s, \xi)$ by the witness-free PPT simulator of Lemma 7.3. By Lemma 7.3, $|\Pr[\mathcal{A}(\mathsf{Hyb}_0^{(b)}) = 1] - \Pr[\mathcal{A}(\mathsf{Hyb}_1^{(b)}) = 1]| \leq \epsilon_{\text{prog}}$.

- $\mathsf{Hyb}_2^{(b)}$: Instead of querying $H_{\text{non}}(\mathsf{enc}(\mathsf{ctx}, R_1, R_2))$, sample $\beta \leftarrow \mathbb{F}_q$ independently without caching that input, retaining the nonaborting condition $R = R_1 + \beta R_2 \neq \mathcal{O}$. For fixed nonidentity R_1, R_2, the map $\beta \mapsto R_1 + \beta R_2$ is a bijection onto $\mathbb{G}$, so R conditioned on $R \neq \mathcal{O}$ is *exactly* uniform over $\mathbb{G}\backslash\{\mathcal{O}\}$, regardless of S_b; there is no additional distributional shift in the value of R. To bound the hidden-input query collision, condition on the final R and on β. If $\beta \neq 0$, the compatible ordered pairs (R_1, R_2) are the $q-2$ equiprobable pairs $(R_1, \beta^{-1}(R - R_1))$ with $R_1 \notin \{\mathcal{O}, R\}$; if $\beta = 0$, $R_1 = R$ and R_2 is independently uniform among the $q-1$ nonidentity group points. Therefore a particular guessed pair has conditional probability at most $1/(q-2)$ in either case. A lazy-sampling/game-hop argument bounds the chance that any of the adversary's $Q_{\text{non}}^{\mathcal{A}}$ queries hits this otherwise hidden input by $Q_{\text{non}}^{\mathcal{A}}/(q-2)$. For reference, an *unconditioned* execution can reject because $R_1 = \mathcal{O}$, $R_2 = \mathcal{O}$, or $R = \mathcal{O}$, with respective bounds $1/(q-1)$, $1/(q-1)$, and $1/q$, giving the conservative sum $3/(q-1)$. This allowance is included in the stated hybrid bound, although R is already exactly uniform under the privacy experiment's nonaborting conditioning. No H_{sig} reprogramming occurs in Hyb_3, so prequeries to that oracle introduce no additional programming conflict. Hence the hybrid distance is at most $Q_{\text{non}}^{\mathcal{A}}/(q-2) + 3/(q-1)$.

- Hyb_3 (*Efficient subset-independent PPT sampler*): Sample scalars $u, v \leftarrow \mathbb{F}_q^{\times}$ independently and uniformly, and set

$$Q_e := uG \in \mathbb{G} \setminus \{\mathcal{O}\}, \qquad R := vG \in \mathbb{G} \setminus \{\mathcal{O}\}.$$

  Query $c \leftarrow H_{\text{sig}}(\mathsf{enc}(\mathsf{CPK}, Q_e, t, R, m))$, compute the Schnorr scalar directly in $\mathbb{F}_q$ as

$$s_e := v + cu \pmod{q},$$

  and run the witness-free PPT simulator of Lemma 7.3 on $\mathsf{st} = \mathsf{enc}(\mathsf{CPK}, n, t, m, Q_e)$ to obtain π_{Auth}. Because $u, v \leftarrow \mathbb{F}_q^{\times}$ induce the exact uniform distribution on $(\mathbb{G} \setminus \{\mathcal{O}\})^2$ and $s_e G = (v + cu)G = R + cQ_e$ is the unique scalar satisfying (10), Proposition 7.2 and Lemma 7.4 imply that Hyb_3 runs in strict polynomial time, is identically distributed to $\mathsf{Hyb}_2^{(b)}$, and is completely independent of b.

Summing the transition bounds across $b \in \{0, 1\}$ via the triangle inequality gives (36). □

8 Concurrent Unauthorized Forgery and the Affine AOMDL Reduction

8.1 Subset-Specific and Message-Level Security Experiments

Definition 8.1 (Subset-specific and message-level concurrent unauthorized forgery)**.** Fix a static set $C \subseteq [n]$ of corrupted registrants with $|C| < t$, registered under the ordered protocol of Assumption 3.1. An adversary $\mathcal{A}$ controls the coordinator, its message and session scheduling, session aborts, and all corrupt signers; it can initiate and interleave up to Q_{sign} honest Round-1 signing sessions in the annotated model of Definition 2.2. Each honest signer executes the one-use state machine of Algorithm 1, including immediate nonce erasure upon Round-2 completion or abort. For each message m, define

$$\mathsf{Resp}(m) = \{\, i \in [n] \setminus C : \text{honest signer } i \text{ released at least one Round-2 scalar on } m \,\}. \quad (37)$$

A released Round-2 scalar counts toward $\mathsf{Resp}(m)$ even if its session is subsequently abandoned by the coordinator.

When $\mathcal{A}$ outputs a candidate forgery (m^*, σ^*) with $\sigma^* = (Q_e^*, R^*, s_e^*, \pi_{\text{Auth}}^*)$ satisfying $\mathsf{Verify}(\mathsf{CPK}, n, t, m^*, \sigma^*) = 1$, let $\mathsf{Ext}(\pi_{\text{Auth}}^*) \mapsto (b^*, d_s^*, \xi^*)$ be the deterministic straight-line authorization extractor of Definition 6.1 that reads $a_L^* = (b^* \,\|\, d_s^* \,\|\, \xi^*)$ from the frozen first-evaluation $\mathcal{D}_{\text{pub}}$-representation (24) of (A^*, S_{com}^*) at $H_y(\mathsf{enc}(T_0^*))$. Recall from Definition 6.1

that extraction succeeds ($\neg\mathsf{Fail}_{\text{ext}}$) if and only if $b^* \in \{0,1\}^n$, $d_s^* \in \{0,1\}^l$, $\sum_{i=1}^n b_i^* - t = \sum_{j=0}^{l-1} 2^j (d_s^*)_j$ in $\mathbb{Z}$ (so $S^* := \{i \in [n] : b_i^* = 1\}$ satisfies $|S^*| \geq t$), and $Q_e^* = \sum_{i=1}^n b_i^* PK_i + \xi^* G$ in $\mathbb{G}$. We define two winning events:

(1) *Subset-specific unauthorized forgery ($\mathsf{Win}_{\mathsf{cuf}}^{\mathsf{sub}}$):* The adversary outputs (m^*, σ^*) accepted by Verify such that either $\mathsf{Ext}(\pi_{\text{Auth}}^*)$ fails ($\mathsf{Fail}_{\text{ext}}$) or $\mathsf{Ext}(\pi_{\text{Auth}}^*)$ succeeds ($\neg\mathsf{Fail}_{\text{ext}}$) with extracted set $S^* = \{i \in [n] : b_i^* = 1\}$ satisfying $S^* \not\subseteq C \cup \mathsf{Resp}(m^*)$:

$$\mathsf{Win}_{\mathsf{cuf}}^{\mathsf{sub}} := \Big\{ \mathsf{Verify}(\mathsf{CPK}, n, t, m^*, \sigma^*) = 1 \\ \wedge \Big(\mathsf{Fail}_{\text{ext}} \ \vee \ (\neg\mathsf{Fail}_{\text{ext}} \wedge S^* \setminus (C \cup \mathsf{Resp}(m^*)) \neq \emptyset) \Big) \Big\}. \tag{38a}$$

(2) *Message-level threshold unauthorized forgery ($\mathsf{Win}_{\mathsf{cuf}}$):* The adversary outputs (m^*, σ^*) accepted by Verify while fewer than $t - |C|$ honest signers across the entire roster responded on m^*:

$$\mathsf{Win}_{\mathsf{cuf}} := \Big\{ \mathsf{Verify}(\mathsf{CPK}, n, t, m^*, \sigma^*) = 1 \ \wedge \ |C \cup \mathsf{Resp}(m^*)| < t \Big\}. \tag{38b}$$

Write $\epsilon_{\text{cuf}}^{\text{sub}} := \Pr[\mathsf{Win}_{\mathsf{cuf}}^{\mathsf{sub}}]$ and $\epsilon_{\text{cuf}} := \Pr[\mathsf{Win}_{\mathsf{cuf}}]$. Because $\neg\mathsf{Fail}_{\text{ext}}$ implies $|S^*| \geq t$, whenever $|C \cup \mathsf{Resp}(m^*)| < t$ and $\neg\mathsf{Fail}_{\text{ext}}$ hold we necessarily have $|S^* \setminus (C \cup \mathsf{Resp}(m^*))| \geq |S^*| - |C \cup \mathsf{Resp}(m^*)| \geq 1$, so $S^* \setminus (C \cup \mathsf{Resp}(m^*)) \neq \emptyset$. Consequently,

$$\mathsf{Win}_{\mathsf{cuf}} \subseteq \mathsf{Win}_{\mathsf{cuf}}^{\mathsf{sub}} \qquad \text{and} \qquad \epsilon_{\text{cuf}} \leq \epsilon_{\text{cuf}}^{\text{sub}}.$$

Remark 8.2 (Interpretation of $\mathsf{Win}_{\mathsf{cuf}}^{\mathsf{sub}}$ vs. $\mathsf{Win}_{\mathsf{cuf}}$). Proving our main security bound directly for the stronger event $\mathsf{Win}_{\mathsf{cuf}}^{\mathsf{sub}}$ in (38a) establishes both security guarantees simultaneously: outside the bounded failure probability, (i) $\mathsf{Ext}(\pi_{\text{Auth}}^*)$ succeeds in extracting a qualified set $S^* \subseteq [n]$ of size $|S^*| \geq t$ linked to Q_e^*, and (ii) *every honest participant $i \in S^* \setminus C$ in that extracted authorization set S^* must have actually released a Round-2 response on message m^*.* As an immediate corollary, at least $|S^* \setminus C| \geq t - |C|$ honest participants on the roster released a Round-2 response on m^* ($\neg\mathsf{Win}_{\mathsf{cuf}}$). Note that this remains an extractor-relative, message-level participation guarantee: it does not assert that all honest members of $S^* \setminus C$ responded within the same coordinator session sid or inspected the final masked key Q_e^*.

Definition 8.3 (M-Algebraic One-More Discrete Logarithm (M-AOMDL))**.** For an integer $M \geq 1$, the challenger samples $w = (w_1, \ldots, w_M)^\top \leftarrow \mathbb{F}_q^M$ uniformly and provides $(W_1, \ldots, W_M) \in \mathbb{G}^M$ with $W_j = w_j G$ to a solver $\mathcal{B}_{\mathsf{AOMDL}}$. The solver has access to an oracle $\mathsf{DL}(\cdot)$ that may be queried at most $M - 1$ times: on input a point $P = a_0 G + \sum_{j=1}^M a_j W_j$ together with a valid linear representation $(a_0, a_1, \ldots, a_M) \in \mathbb{F}_q^{M+1}$, the oracle returns $\log_G(P) = a_0 + \sum_{j=1}^M a_j w_j \pmod q$. The solver wins if it outputs $w \in \mathbb{F}_q^M$. We write $\mathsf{Adv}_{M\text{-}\mathsf{AOMDL}}$ for the maximum winning probability of any solver running within the resource bound of the reduction. The representation $(a_0, \ldots, a_M)$ is strictly internal to the $\mathcal{B}_{\mathsf{AOMDL}} \leftrightarrow \mathsf{DL}$ interface and is never exposed to the simulated signature adversary.

8.2 Complete Public-Atom Dictionary, Checkable Predicates, and Three-World Coupling

Let $H := [n] \setminus C$ be the set of honest participant indices, let Q_{sign} be the upper bound on initiated honest Round-1 sessions across all honest signers, and let $Q_1 \leq Q_{\text{sign}}$ be the number of sessions actually initiated during the execution. Each initiated session is assigned a unique global slot index $s \in [Q_1]$ with honest signer $i_s \in H$, session identifier sid_s, message m_s, immutable context $\mathsf{ctx}_s = \mathsf{enc}(\mathsf{sid}_s, \mathsf{CPK}, t, m_s, Q_{e,s}, \mathsf{enc}(S_s))$, and Round-1 nonce commitments $(U_{s,1}, U_{s,2}) = (r_{s,1} G, r_{s,2} G)$. Because Line 2 and Line 5 of Algorithm 1 reject any repeated sid at honest signer i_s, the pair (i_s, ctx_s) uniquely identifies the global slot $s \in [Q_1]$ (whereas distinct

honest signers participating in the same coordinator session may share the same context string ctx_s). Let $\mathcal{R} \subseteq [Q_1]$ be the subset of slots in which the honest signer releases a Round-2 response s_s, with $Q_2 := |\mathcal{R}| \le Q_1$.

Padding unused session indices $Q_1 < s \le Q_{\text{sign}}$ with independent dummy nonce variables $(r_{s,1}, r_{s,2})$, we collect all $D := |H| + 2Q_{\text{sign}}$ honest secret scalars into the column vector

$$z := \Big((x_i)_{i\in H},\ (r_{s,1}, r_{s,2})_{s\in[Q_{\text{sign}}]}\Big)^\top \in \mathbb{F}_q^D. \tag{39}$$

For each responding slot $s \in \mathcal{R}$, the honest signer computes $\beta_s = H_{\text{non}}(\mathsf{enc}(\mathsf{ctx}_s, R_{s,1}, R_{s,2}))$, verifies $R_s = R_{s,1} + \beta_s R_{s,2} \neq \mathcal{O}$, computes $c_s = H_{\text{sig}}(\mathsf{enc}(\mathsf{CPK}, Q_{e,s}, t, R_s, m_s))$, and releases

$$s_s = \langle q_s, z\rangle \pmod q, \qquad q_s := e_{r_{s,1}} + \beta_s e_{r_{s,2}} + c_s e_{x_{i_s}} \in \mathbb{F}_q^D. \tag{40}$$

Let $\mathbf{Q} \in \mathbb{F}_q^{Q_2\times D}$ denote the response matrix formed by stacking the rows $q_s^\top$ for $s \in \mathcal{R}$. Because the $Q_2 \times Q_2$ submatrix of $\mathbf{Q}$ on the private columns $\{r_{s,1} : s \in \mathcal{R}\}$ is the identity matrix I_{Q_2}, we have unconditionally:

$$\mathsf{rank}(\mathbf{Q}) = Q_2. \tag{41}$$

Official public dictionary and corrupt-key normalization. By Definitions 2.1 and 2.2, the complete public group dictionary available to $\mathcal{A}$ is

$$\mathcal{D}_{\text{pub}} = \Big(G,\ M_1, \ldots, M_d,\ (PK_i)_{i\in[n]},\ (U_{s,1}, U_{s,2})_{s\le Q_1}\Big). \tag{42}$$

Because each corrupted key PK_i ($i \in C$) has a known logarithm $x_i \in \mathbb{F}_q^\times$ extracted during Phase 3 of Assumption 3.1, whenever the extractor inspects a frozen $\mathcal{D}_{\text{pub}}$-annotation it replaces each PK_i ($i \in C$) by $x_i G$, folding its coefficient into the base-G coordinate while leaving the official generator atoms $(M_1, \ldots, M_d)$ and honest atoms $(PK_i)_{i\in H}, (U_{s,1}, U_{s,2})_{s\le Q_1}$ untouched. Because Π_M projects exclusively onto $(M_1, \ldots, M_d)$, this leaves all official-generator projections in Section 6 invariant and ensures that the non-Π_M terms $E_A, E_S, E_{T_1}, E_{T_2}$ in (24) lie strictly in the span of $(G, (PK_i)_{i\in H}, (U_{s,1}, U_{s,2})_{s\le Q_1})$.

For any forgery (m^*, σ^*) with $\sigma^* = (Q_e^*, R^*, s_e^*, \pi_{\text{Auth}}^*)$ satisfying $\mathsf{Win}_{\mathsf{cuf}}^{\mathsf{sub}}$ (38a), let $c^* = H_{\text{sig}}(\mathsf{in}_{\text{sig}}^*)$ where $\mathsf{in}_{\text{sig}}^* := \mathsf{enc}(\mathsf{CPK}, Q_e^*, t, R^*, m^*)$. At the first evaluation of H_{sig} on $\mathsf{in}_{\text{sig}}^*$, the frozen $\mathcal{D}_{\text{pub}}$-representation of R^*—after folding corrupt-key coordinates $(PK_i)_{i\in C}$ into $\gamma^* \in \mathbb{F}_q$—takes the exhaustive form

$$R^* = \gamma^* G + \sum_{k=1}^{d} \eta_k^* M_k + \sum_{i\in H} \lambda_i^* PK_i + \sum_{s=1}^{Q_{\text{sign}}} \left(\alpha_{s,1}^* U_{s,1} + \alpha_{s,2}^* U_{s,2}\right), \tag{43}$$

where $\alpha_{s,1}^* = \alpha_{s,2}^* = 0$ for every slot s not yet initiated when $\mathsf{in}_{\text{sig}}^*$ was first queried.

Simulator-checkable failure predicates ($\mathsf{Fail}_{\mathsf{ext}}$ and $\mathsf{Fail}_{\mathsf{gen}} = G_{\mathbf{auth}} \vee G_{\mathbf{forg}}$). From the frozen representation (24) of (A^*, S_{com}^*) at the first evaluation of $H_y(\mathsf{enc}(T_0^*))$, the straight-line extractor $\mathsf{Ext}(\pi_{\text{Auth}}^*)$ reads $a_L^* = (b^* \,\|\, d_s^* \,\|\, \xi^*) \in \mathbb{F}_q^N$. As in Definition 6.1, $\mathsf{Fail}_{\text{ext}}$ holds if any of the following four conditions fails:

$$b^* \in \{0,1\}^n, \qquad d_s^* \in \{0,1\}^l, \qquad \sum_{i=1}^{n} b_i^* - t = \sum_{j=0}^{l-1} 2^j (d_s^*)_j \ \text{ in } \mathbb{Z}, \qquad Q_e^* = \sum_{i=1}^{n} b_i^* PK_i + \xi^* G \ \text{ in } \mathbb{G}. \tag{44}$$

Whenever $\neg\mathsf{Fail}_{\text{ext}}$ holds, $Q_e^* = \sum_{i=1}^n b_i^* PK_i + \xi^* G$ is a verified identity in $\mathbb{G}$; substituting it together with (43) into the verified Schnorr equation $s_e^* G = R^* + c^* Q_e^*$ (10) yields the exact

zero-group equality in $\mathbb{G}$:

$$\begin{aligned}\mathcal{O} = &\left(s_e^* - \gamma^* - c^*\Big(\xi^* + \sum_{i\in C} b_i^* x_i\Big)\right) G - \sum_{k=1}^{d} \eta_k^* M_k \\ &- \sum_{i\in H} (\lambda_i^* + c^* b_i^*) PK_i - \sum_{s=1}^{Q_{\text{sign}}} (\alpha_{s,1}^* U_{s,1} + \alpha_{s,2}^* U_{s,2}).\end{aligned} \tag{45}$$

Recall from Section 6 that G_{auth} is the checkable event on π^*_{Auth} that $\Pi_M(\Delta_{\text{C1}}) \neq \mathbf{0}$ or $\Pi_M(\Delta_{\text{C4}}) \neq \mathbf{0}$. Define the checkable forgery generator-relation event

$$G_{\text{forg}} := \neg\mathsf{Fail}_{\text{ext}} \ \wedge\ ((\eta_1^*, \ldots, \eta_d^*) \neq \mathbf{0}),$$

and define the combined checkable generator-failure predicate

$$\mathsf{Fail}_{\text{gen}} := G_{\text{auth}} \vee G_{\text{forg}}.$$

Because G_{auth}, $\mathsf{Fail}_{\text{ext}}$, and $(\eta_1^*, \ldots, \eta_d^*)$ are determined directly by the frozen $\mathcal{D}_{\text{pub}}$-annotations and $(PK_i)_{i\in[n]}$, both $\mathsf{Fail}_{\text{ext}}$ and $\mathsf{Fail}_{\text{gen}}$ are efficiently checkable without knowing z.

We specify the standalone reference-world d-DLR / DLP reduction $\mathcal{B}_{\mathsf{DLR}}$ in Algorithm 3 and prove its exact coupling to Game 1.

Algorithm 3: Reference-world generator-relation reduction $\mathcal{B}_{\mathsf{DLR}}(M_1, \ldots, M_d)$

1: **Official generator setup:** Receive challenge generators $(M_1, \ldots, M_d) \in (\mathbb{G} \setminus \{\mathcal{O}\})^d$ (in the random generator model of Definition 2.1(a), $\mathcal{B}_{\mathsf{DLP}}$ receives $Y = uG$, samples $k^* \leftarrow [d]$, sets $M_{k^*} := Y$, and sets $M_k := \gamma_k G$ for $\gamma_k \leftarrow \mathbb{F}_q^\times$, $k \neq k^*$). Publish $M_k = H_{\text{gen}}(\mathfrak{a}_k)$ for $k \in [d]$.

2: **Registration and secret sampling:** Receive static corruption set $C \subseteq [n]$ ($|C| < t$) from $\mathcal{A}$. Sample $z \leftarrow \mathbb{F}_q^D$ uniformly in the clear. Set $PK_i := z_{x_i} G$ ($i \in H$) and $U_{s,k} := z_{r_{s,k}} G$ ($s \in [Q_{\text{sign}}], k \in \{1,2\}$). If $\mathsf{Bad}_{\text{val}}$ occurs (any coordinate of z is 0 or $z_{x_i} = z_{x_j}$ for distinct $i, j \in H$), **abort**. Otherwise publish $(PK_i)_{i\in H}$ and process Phase 3 corrupted keys $(PK_i, x_i)_{i\in C}$ under Assumption 3.1.

3: **Signing and RO simulation:** Lazy-sample $H_y, H_z, H_x, H_{\text{non}}, H_{\text{sig}}$ and record first-evaluation $\mathcal{D}_{\text{pub}}$-annotations (including composite honest-signer propagation) per Definition 2.2. Answer Round-1 queries with $(U_{s,1}, U_{s,2})$ and Round-2 queries with $s_s = \langle q_s, z\rangle \pmod q$ using the known vector z.

4: **Generator-relation extraction:** If $\mathcal{A}$ outputs (m^*, σ^*) satisfying $\mathsf{Verify}(\mathsf{CPK}, n, t, m^*, \sigma^*) = 1$ and $\mathsf{Fail}_{\text{gen}} = G_{\text{auth}} \vee G_{\text{forg}}$ holds:

- Replace all $(PK_i)_{i\in C}$, $(PK_i)_{i\in H}$, and $(U_{s,k})_{s\le Q_1}$ in $\Delta_{\text{C1}}, \Delta_{\text{C4}}$, and (45) by their known base-G multiples using $(x_i)_{i\in C}$ and $z \in \mathbb{F}_q^D$.
- Select the first equation among $\Delta_{\text{C1}} = \mathcal{O}$ (if $\Pi_M(\Delta_{\text{C1}}) \neq \mathbf{0}$), $\Delta_{\text{C4}} = \mathcal{O}$ (if $\Pi_M(\Delta_{\text{C4}}) \neq \mathbf{0}$), and (45) (if G_{forg} holds), and **output** the resulting nontrivial relation $(a, \eta_1, \ldots, \eta_d) \in \mathbb{F}_q^{1+d}$ satisfying $aG + \sum_{k=1}^d \eta_k M_k = \mathcal{O}$ with $(\eta_1, \ldots, \eta_d) \neq \mathbf{0}$ (and if running as $\mathcal{B}_{\mathsf{DLP}}$ with $\eta_{k^*} \neq 0$, output $u = -\eta_{k^*}^{-1}(a + \sum_{k\neq k^*} \eta_k \gamma_k) \pmod q$).

Lemma 8.4 (Reference-world generator-relation reduction and coupling)**.** *Let Game 1 be the unconstrained-sampling experiment for* $\mathsf{Win}^{\mathsf{sub}}_{\mathsf{cuf}}$ *(38a) in which $z \leftarrow \mathbb{F}_q^D$ is sampled uniformly at setup, $PK_i = z_{x_i} G$ ($i \in H$) and $U_{s,k} = z_{r_{s,k}} G$ ($s \in [Q_{sign}], k \in \{1,2\}$), and the experiment aborts at setup if* Bad_{val} *occurs. Then Algorithm 3 ($\mathcal{B}_{\mathsf{DLR}}$) simulates Game 1 identically, and on every*

Game 1 execution where $\mathcal{A}$ outputs an accepting signature triggering $\mathsf{Fail}_{gen} = G_{auth} \vee G_{forg}$, $\mathcal{B}_{\mathsf{DLR}}$ outputs a valid d-DLR solution $(a, \eta_1, \ldots, \eta_d)$. Consequently, in Game 1:

$$\Pr[\mathsf{Win}_1^{\mathsf{sub}} \wedge (\mathsf{Fail}_{ext} \vee \mathsf{Fail}_{gen})] \le \epsilon_{auth} + \mathsf{Adv}_{d\text{-}\mathsf{DLR}} \quad (\le \epsilon_{auth} + d\,\mathsf{Adv}_{\mathsf{DLP}} \text{ under Definition 2.1(a)}). \tag{46}$$

Proof. Steps 1–3 of Algorithm 3 execute the exact Game 1 distribution. Whenever an accepting signature satisfies $\mathsf{Fail}_{\text{gen}} = G_{\text{auth}} \vee G_{\text{forg}}$, either G_{auth} holds (so $\Delta_{\text{C1}} = \mathcal{O}$ or $\Delta_{\text{C4}} = \mathcal{O}$ is a verified group equality with nonzero Π_M-projection) or $\neg G_{\text{auth}} \wedge G_{\text{forg}}$ holds (so $\neg\mathsf{Fail}_{\text{ext}}$ holds by definition of G_{forg}, ensuring that (45) is a verified zero-group equality in $\mathbb{G}$ with $(\eta_1^*, \ldots, \eta_d^*) \neq \mathbf{0}$). Because $\mathcal{B}_{\mathsf{DLR}}$ knows $z \in \mathbb{F}_q^D$ and $(x_i)_{i\in C}$ in the clear and $\mathcal{D}_{\text{pub}}$ contains no auxiliary generators outside $\{M_1, \ldots, M_d\}$, Step 4 deterministically recovers $(a, \eta_1, \ldots, \eta_d)$ with $(\eta_1, \ldots, \eta_d) \neq \mathbf{0}$ such that $aG + \sum_{k=1}^{d} \eta_k M_k = \mathcal{O}$, so $\Pr[\mathsf{Win}_1^{\mathsf{sub}} \wedge \mathsf{Fail}_{\text{gen}}] \le \mathsf{Adv}_{d\text{-}\mathsf{DLR}}$ (which is at most $d\,\mathsf{Adv}_{\mathsf{DLP}}$ under random generators by Lemma 6.2). Finally, since $(\mathsf{Fail}_{\text{ext}} \vee \mathsf{Fail}_{\text{gen}}) = (\mathsf{Fail}_{\text{ext}} \wedge \neg G_{\text{auth}}) \vee \mathsf{Fail}_{\text{gen}}$ and Theorem 6.7 gives $\mathsf{Win}_1^{\mathsf{sub}} \wedge \mathsf{Fail}_{\text{ext}} \wedge \neg G_{\text{auth}} \subseteq \mathsf{Bad}_{\text{auth}}$, a union bound yields (46). □

On every winning execution of $\mathsf{Win}_{\mathsf{cuf}}^{\mathsf{sub}}$ where $\neg\mathsf{Fail}_{\text{ext}} \wedge \neg\mathsf{Fail}_{\text{gen}}$ holds, we have $\neg\mathsf{Fail}_{\text{ext}}$, $S^* \setminus (C \cup \mathsf{Resp}(m^*)) \neq \emptyset$, and $\neg G_{\text{forg}}$, which forces $\eta_1^* = \cdots = \eta_d^* = 0$ in (45). Taking discrete logarithms of (45) base G then yields the forgery linear equation

$$\langle q^*, z\rangle = y^* \pmod q, \qquad y^* := s_e^* - \gamma^* - c^*\Big(\xi^* + \sum_{i\in C} b_i^* x_i\Big) \pmod q, \tag{47}$$

where $q^* \in \mathbb{F}_q^D$ is the explicitly known vector with coordinates

$$q_{x_i}^* = \lambda_i^* + c^* b_i^* \ \ (i \in H), \qquad (q_{r_{s,1}}^*, q_{r_{s,2}}^*) = (\alpha_{s,1}^*, \alpha_{s,2}^*) \ \ (s \in [Q_{\text{sign}}]). \tag{48}$$

8.3 Chronological Two-Nonce Rank Independence and Exhaustive Schedule Audit

Let Q_{non} and Q_{sig} bound the total numbers of distinct first evaluations of H_{non} and H_{sig} (including honest-signer and verification evaluations). Define the checkable rank-dependence failure predicate $\mathsf{Fail}_{\text{dep}}$ to hold on an execution satisfying $\mathsf{Win}_{\mathsf{cuf}}^{\mathsf{sub}}$ (38a) if $\neg\mathsf{Fail}_{\text{ext}} \wedge \neg\mathsf{Fail}_{\text{gen}}$ holds (so $S^* \setminus (C \cup \mathsf{Resp}(m^*)) \neq \emptyset$ and q^* in (48) is well-defined) but $q^* \in \mathsf{rowspan}(\mathbf{Q})$ (equivalently, $\mathsf{rank}\left(\begin{smallmatrix}\mathbf{Q}\\ q^{*\top}\end{smallmatrix}\right) < Q_2 + 1$).

Lemma 8.5 (Chronological rank lemma for subset-specific unforgeability)**.** *In the annotated model of Definition 2.2 (including the composite honest-signer propagation rule of Definition 2.2(i)), on any execution of the subset-specific concurrent unauthorized-forgery game (Definition 8.1, Equation (38a)) under any legal concurrent scheduling of Round-1, Round-2, and random-oracle queries, the checkable rank-failure event Fail_{dep} is contained in an unconditional first-query oracle event Bad_{rank} ($\mathsf{Fail}_{dep} \subseteq \mathsf{Bad}_{rank}$), and thus*

$$q^* \notin \mathsf{rowspan}(\mathbf{Q}), \qquad \mathsf{rank}\begin{pmatrix}\mathbf{Q}\\ q^{*\top}\end{pmatrix} = \mathsf{rank}(\mathbf{Q}) + 1 = Q_2 + 1 \tag{49}$$

holds on every $\mathsf{Win}_{\mathsf{cuf}}^{\mathsf{sub}}$ execution outside $\mathsf{Fail}_{ext} \cup \mathsf{Fail}_{gen} \cup \mathsf{Bad}_{rank}$, where

$$\begin{aligned}\Pr[\mathsf{Fail}_{dep}] &\le \epsilon_{rank} := \Pr[\mathsf{Bad}_{rank}]\\ &\le \frac{Q_{non}(Q_{non}-1)}{2q} + \frac{Q_{sign}Q_{non}Q_{sig}}{q} + \frac{|H|Q_{sig}}{q}.\end{aligned} \tag{50}$$

Proof. Index the distinct first evaluations of H_{sig} chronologically by $j \in [Q_{\text{sig}}]$, occurring at transcript step t_j on input $\mathsf{in}^{(j)} = \mathsf{enc}(\mathsf{CPK}, Q_e^{(j)}, t, R^{(j)}, m^{(j)})$. By Definition 2.2(i) and the linearity of the composite representation rule, whether $\mathsf{in}^{(j)}$ is first evaluated directly by $\mathcal{A}$, by

an honest signer in Line 16 of Algorithm 1 (via $\vec{\gamma}_{R^{(j)}} = \vec{\gamma}_{R_1} + \beta\vec{\gamma}_{R_2}$), or at final verification, the frozen $\mathcal{D}_{\text{pub}}$-representation of $R^{(j)}$ at step t_j uniquely fixes the coefficients $(\lambda_i^{(j)})_{i\in H}$ and $(\alpha_{s,1}^{(j)}, \alpha_{s,2}^{(j)})_{s\in[Q_{\text{sign}}]}$ prior to sampling the output $c_j := H_{\text{sig}}(\mathsf{in}^{(j)}) \leftarrow \mathbb{F}_q$ (with $\alpha_{s,1}^{(j)} = \alpha_{s,2}^{(j)} = 0$ for any slot s not yet initiated at t_j). For each slot $s \in [Q_{\text{sign}}]$ with $\alpha_{s,1}^{(j)} \neq 0$, this fixes a unique target ratio

$$\beta_s^{(j)} := (\alpha_{s,1}^{(j)})^{-1}\alpha_{s,2}^{(j)} \in \mathbb{F}_q.$$

We define three bad events over the chronological query transcript:

- *Event* (N1) *(Nonce-hash collision):* Two distinct first evaluations of H_{non} return the same scalar in $\mathbb{F}_q$. By a union bound over all query pairs, $\Pr[\mathsf{N1}] \le \binom{Q_{\text{non}}}{2}/q$.

- *Event* (N2) *(Late binding-ratio hit):* Any fresh evaluation of H_{non} at step t returns a scalar β equal to $\beta_s^{(j)}$ for some earlier H_{sig} evaluation j ($t_j < t$) and some slot $s \in [Q_{\text{sign}}]$ that had $\alpha_{s,1}^{(j)} \neq 0$ (note that a single ctx string in an H_{non} query may be shared by multiple honest slots s participating in the same coordinator session). At the moment a fresh H_{non} query is evaluated, each prior H_{sig} query j and each slot $s \in [Q_{\text{sign}}]$ with $\alpha_{s,1}^{(j)} \neq 0$ specifies at most one target scalar $\beta_s^{(j)} \in \mathbb{F}_q$, giving at most $Q_{\text{sign}}Q_{\text{sig}}$ target values in total. A union bound over all Q_{non} fresh evaluations yields $\Pr[\mathsf{N2}] \le Q_{\text{sign}}Q_{\text{non}}Q_{\text{sig}}/q$.

- *Event* (S) *(Frozen cross-message signature-hash relation):* Fix a query index $j \in [Q_{\text{sig}}]$ (at step t_j) and an honest participant $i \in H$. Because Line 2 and Line 5 of Algorithm 1 forbid signer i from reusing a session identifier sid, each initiated slot s of signer i ($i_s = i$) has a distinct immutable context $\mathsf{ctx}_s = \mathsf{enc}(\mathsf{sid}_s, \mathsf{CPK}, t, m_s, Q_{e,s}, \mathsf{enc}(S_s))$ that uniquely determines $(m_s, Q_{e,s})$. For every slot s of signer i already initiated prior to t_j with $m_s \neq m^{(j)}$ and $\alpha_{s,1}^{(j)} \neq 0$, inspect the finite set of H_{non} queries of the form $\mathsf{enc}(\mathsf{ctx}_s, R'_{s,1}, R'_{s,2})$ evaluated strictly before step t_j. Outside (N1), at most one such prior H_{non} query returned $\beta_s^{(j)}$. Let $\mathcal{S}_{j,i}^{\text{prior}}$ be the set of slots s of signer i (with $m_s \neq m^{(j)}$ and $\alpha_{s,1}^{(j)} \neq 0$) for which such a matching prior H_{non} query exists at step t_j. For each $s \in \mathcal{S}_{j,i}^{\text{prior}}$, the matching prior query uniquely fixes $(R'_{s,1}, R'_{s,2})$, the point $R'_s = R'_{s,1} + \beta_s^{(j)}R'_{s,2}$, and the candidate H_{sig} input string $h_s^{(j)} := \mathsf{enc}(\mathsf{CPK}, Q_{e,s}, t, R'_s, m_s)$ at step t_j. Grouping $\mathcal{S}_{j,i}^{\text{prior}}$ by distinct candidate strings h defines the net coefficients

$$a_{j,i,h} := \sum_{s\in\mathcal{S}_{j,i}^{\text{prior}}:h_s^{(j)}=h} \alpha_{s,1}^{(j)} \pmod q.$$

  Let $\mathcal{H}_{j,i} := \{\, h : a_{j,i,h} \neq 0 \,\}$. Because every $s \in \mathcal{S}_{j,i}^{\text{prior}}$ has $m_s \neq m^{(j)}$, prefix-free injectivity of enc ensures $\mathsf{in}^{(j)} \notin \mathcal{H}_{j,i}$. Moreover, the set $\mathcal{H}_{j,i}$, the coefficients $(a_{j,i,h})_{h\in\mathcal{H}_{j,i}}$, and $\lambda_i^{(j)}$ are completely determined by the transcript strictly prior to sampling $c_j = H_{\text{sig}}(\mathsf{in}^{(j)})$ at step t_j.

  To bound the probability of the relation

$$c_j - D_{j,i} = 0 \pmod q, \qquad D_{j,i} := \sum_{h\in\mathcal{H}_{j,i}} a_{j,i,h}H_{\text{sig}}(h) - \lambda_i^{(j)} \pmod q, \tag{51}$$

  In particular, when $\mathcal{H}_{j,i} = \emptyset$ (either no matching slots exist or all grouped coefficients cancel), $D_{j,i} = -\lambda_i^{(j)}$ is fixed before c_j is sampled, and the equality has probability $1/q$. we use deferred random-oracle sampling: conditioned on the transcript just before c_j is drawn at step t_j, mentally sample and fix the uniform values $H_{\text{sig}}(h) \leftarrow \mathbb{F}_q$ for any $h \in \mathcal{H}_{j,i}$ not yet queried by step t_j (without revealing them to $\mathcal{A}$ until h is actually queried). This fixes the scalar $D_{j,i} \in \mathbb{F}_q$. Because $\mathsf{in}^{(j)} \notin \mathcal{H}_{j,i}$, the fresh draw $c_j = H_{\text{sig}}(\mathsf{in}^{(j)}) \leftarrow \mathbb{F}_q$ is statistically independent of $D_{j,i}$ ($c_j \perp D_{j,i}$), so $\Pr[c_j = D_{j,i}] = 1/q$. Taking a union bound over all $(j,i) \in [Q_{\text{sig}}] \times H$ gives $\Pr[\mathsf{S}] \le |H|Q_{\text{sig}}/q$.

Set $\mathsf{Bad}_{\mathrm{rank}} := \mathsf{N1} \cup \mathsf{N2} \cup \mathsf{S}$, which satisfies (50). We now prove $\mathsf{Fail}_{\mathrm{dep}} \subseteq \mathsf{Bad}_{\mathrm{rank}}$ across every legal concurrent schedule. Suppose $\mathsf{Fail}_{\mathrm{dep}}$ occurs on an execution satisfying $\mathsf{Win}^{\mathsf{sub}}_{\mathsf{cuf}}$ (38a), and assume toward a contradiction that $\mathsf{Bad}_{\mathrm{rank}}$ did not occur. By definition of $\mathsf{Fail}_{\mathrm{dep}}$, $\neg\mathsf{Fail}_{\mathrm{ext}} \wedge \neg\mathsf{Fail}_{\mathrm{gen}}$ holds, the extracted set $S^* = \{i \in [n] : b_i^* = 1\}$ satisfies $S^* \setminus (C \cup \mathsf{Resp}(m^*)) \neq \emptyset$, and $q^* = \sum_{s\in\mathcal{R}} \mu_s q_s$ for some scalars $\mu_s \in \mathbb{F}_q$. Let $j^* \in [Q_{\mathrm{sig}}]$ be the chronological index of the first evaluation of H_{sig} on $\mathsf{in}^*_{\mathrm{sig}} = \mathsf{in}^{(j^*)}$ (occurring at transcript step t_{j^*}).

First, partition the global slot index space $[Q_{\mathrm{sign}}]$ into four mutually exclusive and collectively exhaustive subsets $[Q_{\mathrm{sign}}] = \mathcal{P}_1 \sqcup \mathcal{P}_2 \sqcup \mathcal{P}_3 \sqcup \mathcal{P}_4$:

Partition $\mathcal{P}_1$: $\mathcal{P}_1 := \{\, s \in [Q_{\mathrm{sign}}] : s \notin \mathcal{R} \,\}$ (*Uninitiated or abandoned Round-1 slots*). By (40), for every $s \notin \mathcal{R}$, every row of $\mathbf{Q}$ has 0 in columns $r_{s,1}$ and $r_{s,2}$. Comparing coordinates in $\{r_{s,1}, r_{s,2}\}$ between q^* and $\sum_{s'\in\mathcal{R}} \mu_{s'} q_{s'}$ forces

$$\alpha^*_{s,1} = \alpha^*_{s,2} = 0 \qquad \forall s \in \mathcal{P}_1.$$

(*Note:* This rules out any schedule in which $\mathcal{A}$ pre-queries H_{non} on an abandoned slot $s \notin \mathcal{R}$ and attempts to include $U_{s,1}$ with $\alpha^*_{s,1} \neq 0$ in R^* while keeping $q^* \in \mathsf{rowspan}(\mathbf{Q})$.)

Partition $\mathcal{P}_2$: $\mathcal{P}_2 := \{\, s \in \mathcal{R} : \alpha^*_{s,1} = 0 \,\}$ (*Completed Round-2 slots unused in R^**). For every $s \in \mathcal{R}$, row $q_s^\top$ is the unique row of $\mathbf{Q}$ with nonzero entries in $(r_{s,1}, r_{s,2})$, having $(q_{s,r_{s,1}}, q_{s,r_{s,2}}) = (1, \beta_s)$. Thus comparing coordinates in $(r_{s,1}, r_{s,2})$ between q^* and $\sum_{s'\in\mathcal{R}} \mu_{s'} q_{s'}$ forces

$$\mu_s = \alpha^*_{s,1}, \qquad \alpha^*_{s,2} = \alpha^*_{s,1}\beta_s \qquad \forall s \in \mathcal{R}. \tag{52}$$

Hence every $s \in \mathcal{P}_2$ has $\mu_s = \alpha^*_{s,1} = 0$ and $\alpha^*_{s,2} = 0$.

Partition $\mathcal{P}_3$: $\mathcal{P}_3 := \{\, s \in \mathcal{R} : \alpha^*_{s,1} \neq 0 \ \wedge\ t_{\mathrm{non},s} > t_{j^*} \,\}$ (*Active completed slots whose actual Round-2 H_{non} input $\mathsf{in}_{non,s} := \mathsf{enc}(\mathsf{ctx}_s, R_{s,1}, R_{s,2})$ was first evaluated after t_{j^*}*). For any $s \in \mathcal{P}_3$, equation (52) implies that the fresh evaluation of $H_{\mathrm{non}}(\mathsf{in}_{\mathrm{non},s})$ at step $t_{\mathrm{non},s} > t_{j^*}$ returned $\beta_s = (\alpha^*_{s,1})^{-1}\alpha^*_{s,2} = \beta_s^{(j^*)}$, which triggers Event (N2). Since $\neg\mathsf{N2}$ holds, $\mathcal{P}_3 = \emptyset$.

Partition $\mathcal{P}_4$: $\mathcal{P}_4 := \{\, s \in \mathcal{R} : \alpha^*_{s,1} \neq 0 \ \wedge\ t_{\mathrm{non},s} < t_{j^*} \,\}$ (*Active completed slots whose actual Round-2 H_{non} input $\mathsf{in}_{non,s}$ was first evaluated before t_{j^*}*). By (52), every $s \in \mathcal{P}_4$ satisfies $\beta_s = \beta_s^{(j^*)}$. Moreover, if $\mathcal{A}$ also queried H_{non} prior to t_{j^*} on any alternative pair $\mathsf{enc}(\mathsf{ctx}_s, R'_{s,1}, R'_{s,2}) \neq \mathsf{in}_{\mathrm{non},s}$, then outside Event (N1) that distinct query cannot also output $\beta_s = \beta_s^{(j^*)}$. Thus $\mathsf{in}_{\mathrm{non},s} = \mathsf{enc}(\mathsf{ctx}_s, R_{s,1}, R_{s,2})$ is the *unique* prior H_{non} query for ctx_s returning $\beta_s^{(j^*)}$, and the candidate string $h_s^{(j^*)}$ recorded in Event (S) at step t_{j^*} is guaranteed to equal the actual Round-2 signing-hash input $h_s = \mathsf{enc}(\mathsf{CPK}, Q_{e,s}, t, R_s, m_s)$ of slot s!

Now pick any honest participant $i^* \in S^* \setminus (C \cup \mathsf{Resp}(m^*)) \subseteq H$. Because $i^* \in S^*$, we have $b^*_{i^*} = 1$; because $i^* \notin \mathsf{Resp}(m^*)$, every completed session $s \in \mathcal{R}$ of signer i^* has $m_s \neq m^*$ (and note that even if $\mathsf{in}^{(j^*)}$ was first evaluated by some honest signer $i_{\tilde{s}}$ in Line 16 of Algorithm 1, Line 18 immediately places $i_{\tilde{s}} \in \mathsf{Resp}(m^*)$, so $i_{\tilde{s}} \neq i^*$). Comparing the x_{i^*}-coordinate of q^* in (48) with $\sum_{s\in\mathcal{R}} \mu_s q_s$ and using $\mathcal{R} = \mathcal{P}_2 \sqcup \mathcal{P}_3 \sqcup \mathcal{P}_4$ with $\mu_s = 0$ on $\mathcal{P}_2$ and $\mathcal{P}_3 = \emptyset$ gives

$$\lambda^*_{i^*} + c^* = \sum_{s\in\mathcal{P}_4 : i_s = i^*} \alpha^*_{s,1} c_s \pmod q. \tag{53}$$

Finally, we verify the bidirectional set equality $\{\, s \in \mathcal{P}_4 : i_s = i^* \,\} = \mathcal{S}^{\mathrm{prior}}_{j^*,i^*}$:

- ($\subseteq$): Every $s \in \mathcal{P}_4$ with $i_s = i^*$ has $m_s \neq m^*$, $\alpha^*_{s,1} \neq 0$, and its actual Round-2 query $\mathsf{in}_{\mathrm{non},s}$ was evaluated at $t_{\mathrm{non},s} < t_{j^*}$ and returned $\beta_s = \beta_s^{(j^*)}$, so $s \in \mathcal{S}^{\mathrm{prior}}_{j^*,i^*}$.

- ($\supseteq$): Conversely, any slot $s \in \mathcal{S}^{\text{prior}}_{j^*,i^*}$ has $i_s = i^*$ and $\alpha^*_{s,1} = \alpha^{(j^*)}_{s,1} \neq 0$. Since $\alpha^*_{s,1} \neq 0$, s cannot lie in $\mathcal{P}_1 \sqcup \mathcal{P}_2$ (where $\alpha^*_{s,1} = 0$), and since $\mathcal{P}_3 = \emptyset$, s must lie in $\mathcal{P}_4$! Furthermore, by Partition $\mathcal{P}_4$, its unique matching prior H_{non} query is $\mathsf{in}_{\text{non},s}$, so $h_s^{(j^*)} = h_s$ and $H_{\text{sig}}(h_s^{(j^*)}) = c_s$.

Regrouping the contributing responses by the distinct candidate H_{sig} inputs $h = h_s^{(j^*)} = h_s$ and cancelling net-zero coefficients gives

$$\begin{aligned}\sum_{\substack{s\in\mathcal{P}_4\\ i_s=i^*}} \alpha^*_{s,1} c_s &= \sum_{s\in\mathcal{S}^{\text{prior}}_{j^*,i^*}} \alpha^{(j^*)}_{s,1} H_{\text{sig}}(h_s^{(j^*)}) \\ &= \sum_{h\in\mathcal{H}_{j^*,i^*}} a_{j^*,i^*,h} H_{\text{sig}}(h) \pmod q.\end{aligned}$$

Substituting into (53) yields $c_{j^*} - D_{j^*,i^*} = 0 \pmod q$, exactly Event (S) in (51). This contradicts $\neg$S; therefore $\mathsf{Fail}_{\text{dep}} \subseteq \mathsf{Bad}_{\text{rank}}$, proving (49) and (50). □

8.4 Opaque Affine Embedding, Deterministic Execution Map, and Projected Rank

Let $M := Q_{\text{sign}} + 1$ and $D := |H| + 2Q_{\text{sign}}$. Given an M-AOMDL challenge exponent vector $w \in \mathbb{F}_q^M$, the reduction samples independent uniform

$$A \leftarrow \mathbb{F}_q^{D\times M}, \qquad b_0 \leftarrow \mathbb{F}_q^D, \qquad \text{and sets} \qquad z := Aw + b_0 \pmod q. \tag{54}$$

Lemma 8.6 (Deterministic execution factorization and opaque-atom transcript independence)**.** *Fix any algebraic adversary $\mathcal{A}$ conforming to Definition 2.2 with internal coin tape $\omega_{\mathcal{A}}$, and let $\mathsf{RO} = (\mathsf{RO}_y, \mathsf{RO}_z, \mathsf{RO}_x, \mathsf{RO}_{non}, \mathsf{RO}_{sig})$ denote the tuple of random-oracle tables.*

(1) Deterministic execution map $\Phi_{\mathcal{A}}$: *Under the syntactic opaque-atom interface of Definition 2.2, there exists a single deterministic function $\Phi_{\mathcal{A}}$ such that in Game 1, in $\mathcal{B}_{\mathsf{DLR}}$ (Algorithm 3), and in $\mathcal{B}_{\mathsf{AOMDL}}$, the complete adversarial view $\mathsf{View}_{\mathcal{A}}$, the frozen $\mathcal{D}_{pub}$-annotation table $\mathcal{F}_{frozen}$, the output (m^*, σ^*), the predicates $\mathsf{Bad}_{val}, \mathsf{Win}^{\mathsf{sub}}_{\mathsf{cuf}}, \mathsf{Fail}_{ext}, G_{auth}, G_{forg}, \mathsf{Fail}_{dep}$, and the augmented matrix $B = \binom{\mathbf{Q}}{q^{*\top}}$ are given deterministically by*

$$(\mathsf{View}_{\mathcal{A}}, \mathcal{F}_{frozen}, m^*, \sigma^*, \mathsf{Bad}_{val}, \mathsf{Win}^{\mathsf{sub}}_{\mathsf{cuf}}, \mathsf{Fail}_{sim}, B) = \Phi_{\mathcal{A}}(\omega_{\mathcal{A}}, \mathbf{M}, z, \mathsf{RO}),$$

where $\mathbf{M} = (M_1, \ldots, M_d)$ is the official generator tuple and $z \in \mathbb{F}_q^D$ is the vector of base-G discrete logarithms of Honest keys $(PK_i)_{i\in H}$ and Round-1 nonces $(U_{s,1}, U_{s,2})_{s\in[Q_{sign}]}$.

(2) Exact distributional alignment and conditional uniformity of A: *For any fixed $w \in \mathbb{F}_q^M$, the affine map (54) induces a marginal distribution on $(\omega_{\mathcal{A}}, \mathbf{M}, z, \mathsf{RO})$—and hence on $\Phi_{\mathcal{A}}(\omega_{\mathcal{A}}, \mathbf{M}, z, \mathsf{RO})$—that is identical to Game 1 and $\mathcal{B}_{\mathsf{DLR}}$, while A remains conditionally uniform on $\mathbb{F}_q^{D\times M}$ given $(\omega_{\mathcal{A}}, \mathbf{M}, z, \mathsf{RO})$:*

$$\mathcal{L}(A \mid \omega_{\mathcal{A}}, \mathbf{M}, z, \mathsf{RO}, \Phi_{\mathcal{A}}(\omega_{\mathcal{A}}, \mathbf{M}, z, \mathsf{RO})) = \mathcal{L}(A) = \mathsf{Unif}(\mathbb{F}_q^{D\times M}).$$

Proof. For claim (1), observe by induction on the query steps that $\mathcal{A}$ receives at setup only $(G, \mathbf{M})$ and $(PK_i = z_{x_i}G)_{i\in H}$ (after $\neg\mathsf{Bad}_{\text{val}}(z)$ is checked), submits $(PK_i, x_i)_{i\in C}$ as a deterministic function of its fixed random tape and the public transcript up to registration (and hence of $(\omega_{\mathcal{A}}, \mathbf{M}, z, \mathsf{RO})$), with duplicates checked against $\{z_{x_i}G : i \in H\}$), receives $(U_{s,1}, U_{s,2}) = (z_{r_{s,1}}G, z_{r_{s,2}}G)$ on the s-th Round-1 query, receives scalars from RO, and on each valid Round-2 query $s \in \mathcal{R}$ receives $s_s = \langle q_s, z\rangle \pmod q$ (computed directly from z in Game 1 and $\mathcal{B}_{\mathsf{DLR}}$, and returned by $\mathsf{DL}(P_s)$ as $\log_G(P_s) = \langle q_s, Aw + b_0\rangle = \langle q_s, z\rangle \pmod q$ in $\mathcal{B}_{\mathsf{AOMDL}}$).

By Definition 2.2(ii), neither (A, b_0) nor $(W_1, \ldots, W_M)$ is ever written to $\mathcal{A}$'s input tape, so $\mathcal{A}$'s next message and $\mathcal{D}_{\text{pub}}$-annotations depend solely on $(\omega_{\mathcal{A}}, \mathbf{M}, z, \mathsf{RO})$. At the end of the run, $\mathsf{Fail}_{\text{ext}}, G_{\text{auth}}, G_{\text{forg}}, \mathsf{Fail}_{\text{dep}}$, and $B = \binom{\mathbf{Q}}{q^{*\top}}$ are computed deterministically from $\mathcal{F}_{\text{frozen}}, \mathcal{D}_{\text{pub}}$, and $(x_i)_{i \in C}$.

For claim (2), for every fixed (A, w), the map $b_0 \mapsto Aw + b_0 \pmod q$ is a bijection on $\mathbb{F}_q^D$, so $\Pr[z = z_0 \mid A, w] = q^{-D}$. Because $(\omega_{\mathcal{A}}, \mathbf{M}, \mathsf{RO})$ are sampled independently of (A, b_0), the joint law satisfies $\Pr[A = A_0, z = z_0 \mid \omega_{\mathcal{A}}, \mathbf{M}, \mathsf{RO}, w] = q^{-DM} \cdot q^{-D}$, proving both the marginal uniformity of z and the conditional uniformity of A given $(\omega_{\mathcal{A}}, \mathbf{M}, z, \mathsf{RO})$ and its deterministic image $\Phi_{\mathcal{A}}(\omega_{\mathcal{A}}, \mathbf{M}, z, \mathsf{RO})$. □

Lemma 8.7 (Projected-rank preservation, including adaptive row count)**.** *Under the hypotheses of Lemma 8.6, let E be any event determined by the execution transcript, and let B be a transcript-determined matrix with r rows and full row rank $r \le M$ on E. The row count r may depend on the transcript. Then*

$$\Pr\left[E \wedge (\operatorname{rank}(BA) < r)\right] = \mathbb{E}\left[\mathbf{1}_E \left(1 - \prod_{j=0}^{r-1}(1 - q^{-(M-j)})\right)\right] \le \frac{\Pr[E]}{q-1} \le \frac{1}{q-1}. \tag{55}$$

When r is constant on E, the expectation simplifies to $\Pr[E]$ times the displayed rank-deficiency factor. In particular the bound is meaningful even when $\Pr[E] = 0$.

Proof. Condition on the full reference execution tuple $(\omega_{\mathcal{A}}, \mathbf{M}, z, \mathsf{RO})$ at a point of E. This fixes r and a full-row-rank B, whereas Lemma 8.6 leaves the affine matrix A uniform over $\mathbb{F}_q^{D \times M}$. Hence BA is uniform in $\mathbb{F}_q^{r \times M}$, and its deficient-rank probability is $1 - \prod_{j=0}^{r-1}(1 - q^{-(M-j)}) \le \sum_{j=0}^{r-1} q^{-(M-j)} \le 1/(q-1)$. Integrating over all reference executions in E, including every possible value of the random row count r, proves the claimed joint-event bound without conditioning on an event of possibly zero probability. □

8.5 Reduction, Oracle Budget, and Security Bound for $\mathsf{Win}_{\mathsf{cuf}}^{\mathsf{sub}}$ and $\mathsf{Win}_{\mathsf{cuf}}$

Theorem 8.8 (Conditional subset-specific and message-level concurrent unforgeability)**.** *In the model of Definitions 2.1 and 2.2 and Assumption 3.1, assume the parameter bound (2). Suppose an adversary $\mathcal{A}$ corrupts $|C| < t$ participants, initiates at most Q_{sign} honest Round-1 signing sessions, and triggers at most $Q_y, Q_z, Q_x, Q_{non}, Q_{sig}$ distinct first evaluations of $H_y, H_z, H_x, H_{non}, H_{sig}$ (including honest-signer and verification queries). Then for $M = Q_{sign} + 1$, both the message-level forgery probability $\epsilon_{cuf} = \Pr[\mathsf{Win}_{\mathsf{cuf}}]$ in (38b) and the subset-specific forgery probability $\epsilon_{cuf}^{sub} = \Pr[\mathsf{Win}_{\mathsf{cuf}}^{\mathsf{sub}}]$ in (38a) satisfy*

$$\begin{aligned}\epsilon_{cuf} \le \epsilon_{cuf}^{sub} \le \epsilon_{auth} + \mathsf{Adv}_{d\text{-}\mathsf{DLR}} + \mathsf{Adv}_{M\text{-}\mathsf{AOMDL}} + \epsilon_{rank} \\ + \frac{|H|(|H|+1)/2 + 2Q_{sign}}{q} + \frac{1}{q-1}.\end{aligned} \tag{56}$$

where ϵ_{auth} is given in (30), ϵ_{rank} is given in (50), and under the ideal random generator model of Definition 2.1(a) we have $\mathsf{Adv}_{d\text{-}\mathsf{DLR}} \le d\,\mathsf{Adv}_{\mathsf{DLP}}$.

Proof. By Definition 8.1, $\mathsf{Win}_{\mathsf{cuf}} \subseteq \mathsf{Win}_{\mathsf{cuf}}^{\mathsf{sub}}$, so it suffices to bound $\epsilon_{\text{cuf}}^{\text{sub}} = \Pr[\mathsf{Win}_0^{\mathsf{sub}}]$ in the real experiment (Game 0) for $\mathsf{Win}_{\mathsf{cuf}}^{\mathsf{sub}}$ (38a).

Game 0 to Game 1 (Probability coupling for unconstrained secret sampling). Couple Game 0 and Game 1 by sampling $z \leftarrow \mathbb{F}_q^D$ uniformly at the start of Phase 2 of Assumption 3.1 and defining $\mathsf{Bad}_{\text{val}}(z)$ as the event that any coordinate of z is $0 \pmod q$ or $z_{x_i} = z_{x_j}$ for distinct $i, j \in H$. Game 1 aborts if $\mathsf{Bad}_{\text{val}}(z)$ occurs, and otherwise executes $\Phi_{\mathcal{A}}(\omega_{\mathcal{A}}, \mathbf{M}, z, \mathsf{RO})$ identically to Game 0, giving

$$|\Pr[\mathsf{Win}_0^{\mathsf{sub}}] - \Pr[\mathsf{Win}_1^{\mathsf{sub}}]| \le \Pr[\mathsf{Bad}_{\text{val}}] \le \frac{|H|(|H|+1)/2 + 2Q_{\text{sign}}}{q}. \tag{57}$$

Game 1 to Game 2 (Exact partition on $\Phi_{\mathcal{A}}(\omega_{\mathcal{A}}, \mathbf{M}, z, \mathsf{RO})$). On the probability space $(\omega_{\mathcal{A}}, \mathbf{M}, z, \mathsf{RO})$ of Game 1, define $\mathsf{Fail}_{\mathrm{sim}} := \mathsf{Fail}_{\mathrm{ext}} \vee \mathsf{Fail}_{\mathrm{gen}} \vee \mathsf{Fail}_{\mathrm{dep}}$ and $\mathsf{Win}_2^{\mathsf{sub}} := \mathsf{Win}_1^{\mathsf{sub}} \wedge \neg\mathsf{Fail}_{\mathrm{sim}}$. By Lemma 8.6(1), both $\mathcal{B}_{\mathsf{DLR}}$ (Algorithm 3) and Game 1 induce the exact same distribution on $\Phi_{\mathcal{A}}(\omega_{\mathcal{A}}, \mathbf{M}, z, \mathsf{RO})$. Therefore, Lemma 8.4 (46) and Lemma 8.5 (50) ($\mathsf{Fail}_{\mathrm{dep}} \subseteq \mathsf{Bad}_{\mathrm{rank}}$) apply directly on Game 1's probability space to give

$$\begin{aligned}\Pr[\mathsf{Win}_1^{\mathsf{sub}}] - \Pr[\mathsf{Win}_2^{\mathsf{sub}}] &= \Pr[\mathsf{Win}_1^{\mathsf{sub}} \wedge \mathsf{Fail}_{\mathrm{sim}}] \\ &\le \Pr[\mathsf{Win}_1^{\mathsf{sub}} \wedge (\mathsf{Fail}_{\mathrm{ext}} \vee \mathsf{Fail}_{\mathrm{gen}})] + \Pr[\mathsf{Fail}_{\mathrm{dep}}] \\ &\le \epsilon_{\mathrm{auth}} + \mathsf{Adv}_{d\text{-}\mathsf{DLR}} + \epsilon_{\mathrm{rank}}.\end{aligned}$$

On every tuple $(\omega_{\mathcal{A}}, \mathbf{M}, z, \mathsf{RO}) \in \mathsf{Win}_2^{\mathsf{sub}}$, $\neg\mathsf{Fail}_{\mathrm{ext}} \wedge S^* \setminus (C \cup \mathsf{Resp}(m^*)) \neq \emptyset \wedge \neg\mathsf{Fail}_{\mathrm{gen}} \wedge \neg\mathsf{Fail}_{\mathrm{dep}}$ holds by definition, so equations (40), (47), and (49) yield the full-row-rank linear system

$$\begin{aligned} Bz = y_{\mathrm{aug}} \pmod q, \qquad \mathsf{rank}(B) = Q_2 + 1, \\ B := \begin{pmatrix} \mathbf{Q} \\ q^{*\top} \end{pmatrix} \in \mathbb{F}_q^{(Q_2+1)\times D}, \qquad y_{\mathrm{aug}} := \begin{pmatrix} (s_s)_{s\in\mathcal{R}} \\ y^* \end{pmatrix}. \end{aligned} \tag{58}$$

Explicit M-AOMDL reduction $\mathcal{B}_{\mathsf{AOMDL}}$. Given an M-AOMDL challenge $(W_1, \ldots, W_M) = (w_1 G, \ldots, w_M G)$ for $M = Q_{\mathrm{sign}}+1$, $\mathcal{B}_{\mathsf{AOMDL}}$ initializes the official generators $\mathbf{M} = (M_1, \ldots, M_d)$ per Definition 2.1, receives $C \subseteq [n]$ in Phase 1 of Assumption 3.1, samples $A \leftarrow \mathbb{F}_q^{D\times M}$ and $b_0 \leftarrow \mathbb{F}_q^D$ as in (54), and computes $PK_i := b_{0,x_i} G + \sum_{j=1}^M A_{x_i,j} W_j$ $(i \in H)$ and $U_{s,k} := b_{0,r_{s,k}} G + \sum_{j=1}^M A_{r_{s,k},j} W_j$ $(s \in [Q_{\mathrm{sign}}], k \in \{1,2\})$, aborting if $\mathsf{Bad}_{\mathrm{val}}(z)$ occurs. It answers each Round-2 query $s \in \mathcal{R}$ via one call to $\mathsf{DL}(P_s)$ where $P_s = U_{s,1} + \beta_s U_{s,2} + c_s PK_{i_s} = (q_s^\top b_0)G + \sum_{j=1}^M (q_s^\top A)_j W_j$. When $\mathcal{A}$ outputs (m^*, σ^*), the reduction does *not* evaluate the analysis-only map $\Phi_{\mathcal{A}}(\omega_{\mathcal{A}}, \mathbf{M}, z, \mathsf{RO})$ and never attempts to recover the unknown vector z. Instead it executes the explicit public-transcript decision procedure below, using only its recorded responses, first-evaluation annotations, the public roster, and the extracted corrupted-key exponents:

(i) Check public $\mathsf{Verify}(\mathsf{CPK}, n, t, m^*, \sigma^*) = 1$. The setup rejection predicate $\mathsf{Bad}_{\mathrm{val}}$ was already evaluated at initialization using the public points PK_i and $U_{s,k}$: tests $z_\nu = 0$ use equality of the associated point to $\mathcal{O}$, and $z_{x_i} = z_{x_j}$ uses equality of public points. No scalar z_ν is needed.

(ii) Recover $a_L^* = (b^* \| d_s^* \| \xi^*)$ from the frozen annotated A^* and test $\mathsf{Fail}_{\mathrm{ext}}$ using Booleanity, the integer threshold identity, and the *public* point equation $Q_e^* = \sum_i b_i^* PK_i + \xi^* G$. If extraction succeeds, compute $S^* = \{i : b_i^* = 1\}$ and inspect the locally recorded message-indexed set $\mathsf{Resp}(m^*)$. Test the subset-specific winning condition $S^* \setminus (C \cup \mathsf{Resp}(m^*)) \neq \emptyset$. An accepted proof with failed extraction also satisfies the defined winning event, but is rejected at the next check.

(iii) Compute G_{auth} by projecting the frozen annotated zero-group checks (C1),(C4) onto the official generator coordinates; compute G_{forg} by testing whether the official-generator coefficient vector $(\eta_k^*)_{k=1}^d$ of frozen R^* is nonzero, conditional on successful extraction. These are coordinate operations, *not* discrete-logarithm computations.

(iv) If extraction and generator checks pass, calculate q^* from the frozen public-basis coefficients by (48), and Q from the recorded (i_s, β_s, c_s) for each responding slot via (40). Test $\mathsf{Fail}_{\mathrm{dep}}$ by Gaussian elimination over $\mathbb{F}_q$ on $\left(\begin{smallmatrix} Q \\ q^{*\top} \end{smallmatrix}\right)$.

The reduction aborts unless the output is accepted, the subset-specific event holds, and all three $\mathsf{Fail}_{\mathrm{ext}}, \mathsf{Fail}_{\mathrm{gen}}, \mathsf{Fail}_{\mathrm{dep}}$ are false. All checks are polynomial-time and require no knowledge of z. Otherwise $\mathsf{Win}_2^{\mathsf{sub}}$ has occurred and

$$(BA)w = y_{\mathrm{aug}} - Bb_0 \pmod q. \tag{59}$$

By Lemma 8.6(2), $\Pr_{\mathcal{B}_{\mathsf{AOMDL}}}[\mathsf{Win}_2^{\mathsf{sub}}] = \Pr_{\text{Game 1}}[\mathsf{Win}_2^{\mathsf{sub}}]$. Applying Lemma 8.7 to $E = \mathsf{Win}_2^{\mathsf{sub}}$, on which $r = Q_2 + 1 \le M$ (possibly differing between executions), yields the unconditional joint-event bound

$$\Pr_{\mathcal{B}_{\mathsf{AOMDL}}}[\mathsf{Win}_2^{\mathsf{sub}} \wedge \operatorname{rank}(BA) < r] \le \frac{\Pr_{\mathcal{B}_{\mathsf{AOMDL}}}[\mathsf{Win}_2^{\mathsf{sub}}]}{q-1} \le \frac{1}{q-1}.$$

On $\mathsf{Win}_2^{\mathsf{sub}} \wedge (\operatorname{rank}(BA) = r)$, the reduction queries $\mathsf{DL}(W_{j\ell})$ on $M - r = M - Q_2 - 1$ basis-extending rows, making exactly $Q_2 + (M - Q_2 - 1) = M - 1$ discrete-log-oracle queries altogether, and solves the resulting full-rank linear system for the challenge vector w. Thus

$$\begin{aligned}
\mathsf{Adv}_{M\text{-}\mathsf{AOMDL}} &\ge \Pr_{\mathcal{B}_{\mathsf{AOMDL}}}[\mathsf{Win}_2^{\mathsf{sub}} \wedge (\operatorname{rank}(BA) = r)] \\
&= \Pr_{\text{Game 1}}[\mathsf{Win}_2^{\mathsf{sub}}] - \Pr_{\mathcal{B}_{\mathsf{AOMDL}}}[\mathsf{Win}_2^{\mathsf{sub}} \wedge (\operatorname{rank}(BA) < r)] \\
&\ge \Pr_{\text{Game 1}}[\mathsf{Win}_2^{\mathsf{sub}}] - \frac{1}{q-1}.
\end{aligned}$$

Hence $\Pr_{\text{Game 1}}[\mathsf{Win}_2^{\mathsf{sub}}] \le \mathsf{Adv}_{M\text{-}\mathsf{AOMDL}} + 1/(q-1)$. Combining the transition bounds across Games 0–2 yields (56). □

9 Security Scope and Limitations

The completeness claim holds for valid honest nonaborting executions. Authorization knowledge soundness and exact key linkage use the special first-evaluation representation-annotated algebraic group model, which is stronger than standard AGM+ROM. The unforgeability analysis is extractor-relative, message-level, and restricted to static corruption and ideal extractable registration. It does not prove same-session agreement, approval of coordinator-chosen Q_e, adaptive-corruption security, or full interactive-transcript signer privacy. Fixed public hash-to-curve generators require the separately stated d-DLR hardness assumption; no automatic reduction from fixed-parameter DLR to DLP is claimed. The updates to the oracle-order and projected-rank bookkeeping do not establish security outside the stated idealized interfaces. The compiled integration is not by itself a mathematical proof audit; the proofs remain candidates for independent cryptographic verification and are not an implementation security guarantee.

References

[1] C.-P. Schnorr, "Efficient Signature Generation by Smart Cards," *Journal of Cryptology*, 4(3), 1991.

[2] S. Micali, K. Ohta, and L. Reyzin, "Accountable-Subgroup Multisignatures," *ACM CCS*, 2001.

[3] T. Ristenpart and S. Yilek, "The Power of Proofs-of-Possession: Securing Multiparty Signatures against Rogue-Key Attacks," *EUROCRYPT*, 2007.

[4] B. Bünz, J. Bootle, D. Boneh, A. Poelstra, P. Wuille, and G. Maxwell, "Bulletproofs: Short Proofs for Confidential Transactions and More," *IEEE S&P*, 2018.

[5] C. Komlo and I. Goldberg, "FROST: Flexible Round-Optimized Schnorr Threshold Signatures," *SAC*, 2020.

[6] F. Benhamouda, T. Lepoint, J. Loss, M. Orrù, and M. Raykova, "On the (In)security of ROS," *EUROCRYPT*, 2021.

[7] J. Nick, T. Ruffing, and Y. Seurin, "MuSig2: Simple Two-Round Schnorr Multi-Signatures," *CRYPTO*, 2021.

[8] D. Boneh, M. Drijvers, and G. Neven, "Compact Multi-Signatures for Smaller Blockchains," *ASIACRYPT*, 2018.